Informatica® Cloud Data Governance and Catalog

2026 July

Understanding Technical Assets

Informatica Cloud Data Governance and Catalog Understanding Technical Assets
2026 July

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Preface

Read *Understanding Technical Assets* to familiarize yourself with the assets from various source systems for which you can extract metadata to Data Governance and Catalog.

CHAPTER 1

Technical assets

Technical assets are representations of data elements, technical data sets, and data containers that Metadata Command Center extracts from a source system.

The technical assets that you see in Data Governance and Catalog depend on the configurations that your organization administrator applies in Metadata Command Center. These configurations include the metadata extraction settings for the catalog sources, the capabilities that are enabled, the profiling filters that are applied, the job schedules that are defined, and other factors. For more information about catalog source configurations, see *Administration* in the Metadata Command Center help.

Technical assets in Data Governance and Catalog can be broadly categorized in the following manner:

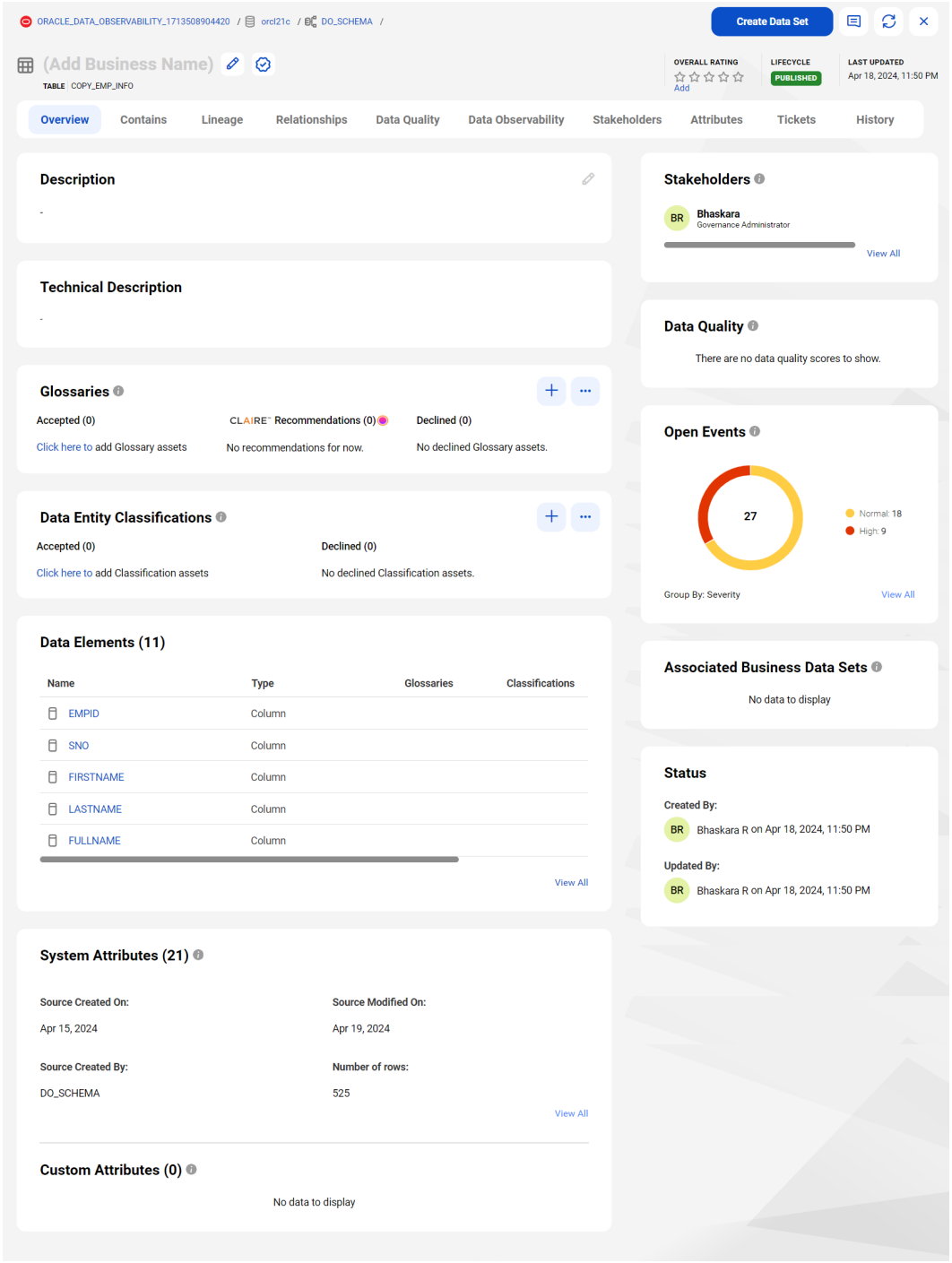
- Data elements are the smallest unit of data in a source system. Columns, files, fields are examples of data elements.
- Technical data sets are a collection of data elements. For example, a table is a technical data set as it is a collection of columns.
- Data containers hold the data for an application. Examples of data containers are schema and database.
- Catalog sources are objects that represent the source system from which you extract metadata. You can create catalog sources in Metadata Command Center to extract metadata from source systems like Amazon S3 and Oracle.

View technical assets

You can view technical asset pages for data elements, technical data sets, data containers, and catalog sources.

The assets that you can view in Data Governance and Catalog depend on the permissions that your organization administrator assigns to you through access policies in Metadata Command Center. For information about how administrators manage access to assets, see the *Asset Management* help.

The following image shows the technical asset page for a sample table in an Oracle catalog source:



The header of the technical asset page displays the following properties of the technical asset:

Field	Description
Hierarchy	The top of the page displays the hierarchical path of the technical asset.
Name	Name of the technical asset. The name can be one of the following types: - Technical name of the asset that Metadata Command Center extracts from the source system. For more information about configuring a catalog source to extract metadata from source systems, see the *Catalog Source Configuration* help in Metadata Command Center. - Readable business name that a user has manually associated with the technical asset or is automatically assigned to the technical asset. For more information about curating glossaries for technical assets, see "Curate glossaries for technical assets" on page 21.
Overall Rating	Rate a technical asset to indicate your assessment of the quality of the asset. As a stakeholder, you can endorse a technical asset as trusted and reliable by certifying the asset and also decertify the technical asset if it is longer trusted and valid. You can add your comments to the asset and collaborate with other users to make collective decisions on using the data. For more information about collaborating on an asset, see *Collaboration* in the *Asset Management* help.
Lifecycle	Stage of the asset in the production lifecycle. If the administrator has configured a workflow for creating or updating assets, the **Lifecycle** value displays the current stage of the asset in the workflow.
Last Updated	Date on which the asset was last updated.

Overview tab

View the basic information about the technical asset. Depending on the type of technical asset that you are viewing, the following table describes the fields that appear on the **Overview** tab:

Field	Description
Description	This field appears only for technical data sets and data elements. Description of the technical asset. The size of the field should not exceed 1 MB. The description can be one of the following types: - Description that is assigned to a technical asset either by its association with an existing glossary business term or by typing a new description. For more information about assigning a business name and description to a technical asset, see the *Asset Management* help. - Description of an associated Glossary asset that is automatically assigned. For more information about configuring automatic association of technical assets with Glossary assets, see the *Asset Management* help.
Technical Description	Technical description of the asset that Metadata Command Center extracts from the source system. This field appears only for technical data sets and data elements. For more information about configuring a catalog source to extract metadata from source systems, see the *Catalog Source Configuration* help in Metadata Command Center.

Field	Description
Glossaries	View glossaries for the technical assets. Accept or decline glossaries that the CLAIRE engine recommends, or associate a glossary asset manually. You can hover over a glossary to view the confidence score based on which the glossary asset was recommended, accepted or rejected. **Note:** If a glossary asset was manually associated with the technical asset, the system doesn't display a confidence score when you hover over the glossary. This field appears only for technical data sets and data elements. For more information, see "Glossary association" on page 21.
Data Element Classifications	View data element classifications for the technical assets. Accept or decline data element classifications that the data classification rule detects, or associate a data element classification manually. For more information, see "Curate data classifications for technical assets" on page 28. This field appears only for technical data elements *only* if the Data Classification capability is enabled and configured in Metadata Command Center for the catalog source.
Data Entity Classifications	View data entity classifications for the technical assets. Accept or decline data entity classifications that the data entity rule detects, or associate a data entity classification manually. For more information, see "Curate data classifications for technical assets" on page 28. This field appears only for technical data sets *only* if the Data Classification capability is enabled and configured in Metadata Command Center for the catalog source.
Document Classifications	View document classifications for the technical assets. Accept or decline document classifications that the document classification capability detects, or manually associate a document classification with the asset. For more information, see "Curate document classifications for technical assets" on page 31. This field appears only for unstructured data assets such as PDFs, text files, and Microsoft Word files.
System Attributes	The system properties of the technical asset.
Custom Attributes	The user configured custom attributes of the technical asset. To view all the existing custom attributes, click **View All** to open the **Custom Attributes** sub-tab on the **Attributes** tab. You can also modify values of the custom attributes from the **Overview** tab or the **Attributes** tab.
Profiling statistics	To know more about the types of profiling statistics that you can view, see "Data profiling statistics" on page 19. This field appears for technical data elements *only* if the Data Profiling capability is enabled and configured in Metadata Command Center for the catalog source.
Data Elements	The child assets of a data set. This section displays the data elements, such as columns, that constitute the data set. You can click a data element to see the details on a new technical asset page. This field appears only for technical data sets.
Data Quality Score	Aggregated data quality scores for the technical asset. The percentage value displays the overall score for the data elements that comprise the asset. The **Rules Against Target** section display the aggregate quality measurement of the data sets and data elements against the defined threshold and target values. Click **View rules against target breakdown** to see the data quality rules that specify the threshold and target values. This field appears only for technical data sets and data elements.

Field	Description
Data Marketplace	Details of an asset's presence in Data Marketplace. If an asset is added to a data collection in Data Marketplace, you can view the details of the data collection. For more information, see the following topics: - To connect Data Governance and Catalog to Data Marketplace, see the *Set Up Data Marketplace* help in Data Marketplace. - To add an asset to a data collection, see the *Working With Data Collections* help in Data Marketplace.
Open Events	Open events for the data observability anomalies identified for the technical asset.
Stakeholders	Users responsible for the technical asset or interested in the asset.
Asset Groups	One or more asset groups assigned to the asset. Asset groups allow users access to a set of assets in the organization. You can assign asset groups to an asset if you are the stakeholder of the asset or if your organization administrator has granted you the **Manage Access** permission on the asset through access policies in Metadata Command Center.
Associated Business Data Sets	The business data sets that are associated with the technical data set. This field appears only for technical data sets and data elements. For information about creating data sets from technical assets, see the *Asset Management* help.
Reference Data	Code values and attributes from Reference 360 linked to the data element. The **Reference Data** field appears only for data elements, and you can view the **Reference Data** tab only if you have the **View Reference Data** permission.
Created By	User that created the technical asset along with the date and time of creation.
Updated By	User that last updated the technical asset along with the date and time on which it was last updated.
Linked Lineage	The links that are generated for this catalog source as a result of linking it with another catalog source in Metadata Command Center. For more information about linking catalog sources to generate lineage, see the *Administration* help in Metadata Command Center. You can view the links for a catalog source only if you are a stakeholder for the catalog source. Click this link to view all the links generated for this catalog source and to curate those links to view complete lineage. For more information about curating the catalog source links, see the *Asset Management* help. This panel appears only for catalog sources.

Note: The **Overview** tab for the following catalog sources display Primary key (PK) and Foreign key (FK) columns:

- SAP ERP
- IDERA ER/Studio Data Architect

Lineage tab

View the data lineage of the technical asset. A lineage graphically displays the inflow of data from other systems into the asset and the outflow of data from the asset to other systems.

For more information about how to view data lineage on this page, see the *Asset Discovery* help.

Contains tab

This tab is displayed for technical data sets. On this tab, the **Data Elements** grid list the data elements that constitute the data set and the data element properties. You can expand the assets on the grid to view the complete hierarchy of all child assets. You can search for top-level assets by their name, description, technical description, and business name. You can also filter the grid by asset type and sort assets based on name, business name, or position.

Click **Add** to add data elements to the data set or create a manual data element within the data set. You can add up to 500 data elements to a data set, for which Data Governance and Catalog triggers a job to track its progress. When you add up to 50 data elements, you add data elements through a faster job. You can track the progress of the jobs on the **Job Monitoring** page. If you want to remove a data element from the data set, hover over it and click **Delete**. For more information about the monitoring jobs, see *Monitor jobs* in the *Asset Discovery* help.

Note: If a job to modify the data elements in the data set is in progress, wait for the job to complete before you add or delete data elements.

The following table describes the properties of data elements that you can view on the **Contains** tab:

Field	Description
Name	Name of the constituent data element. Shows if the constituent data element is part of a Primary or Secondary key.
Business Name	Business name of the data element.
Type	Type of the constituent data element.
Glossaries	Glossary assets that are associated with the constituent data elements. You can use the **Curate** button to associate a data element with glossary assets. You can hover over a glossary to view the confidence score based on which the glossary asset was recommended, accepted or rejected. **Note:** If a glossary asset was manually associated with the technical asset, the system doesn't display a confidence score when you hover over the glossary. For more information, see "Glossary association" on page 21.
Classifications	Data classifications associated with the constituent data elements. You can use the **Curate** button to associate a data element with data classifications. For more information, see "Curate data classifications for technical assets" on page 28.
CLAIRE Recommendations	Automatically recommended glossary assets for the data element.
Profiling statistics	Null, distinct, and non-distinct values for the data element. For more information, see "Data profiling statistics" on page 19.
Source Data Type	Source data type of the data element.
Inferred Data Type	The data type that was inferred for the data element.
Description	Description of the constituent data element as it appears in the source system.
Technical Description	Technical description of the constituent data element as it appears in the source system.
Sensitivity	Sensitivity level of the constituent data element.

Field	Description
Hierarchy	Hierarchy of the constituent data element.
Custom Attributes	Custom attributes of the constituent data element.

Relationships tab

On this tab, you can switch between grid view or graph view to view all the relationships that the technical asset has with other assets. Create relationships with other assets if they can be used together to make business or governance decisions.

The following table describes the fields that appear on the **Relationships** tab:

Field	Description
Name	Name of the related asset.
Type	Type of the related asset.
How It Is Related	Relationship between the assets and the technical asset that you have opened. Hover the mouse over the arrows to see if the relationship is a direct, indirect, inbound or outbound. The term "this asset" refers to the technical asset that is open.

The following table describes the filters you can apply to the list of relationships:

Filter	Description
Asset Type	Select one or more values to view related assets of specific asset types.
Defined Group	Specify whether you want to view related business or technical assets. You can select one of the following values: - Governance: View assets for which the relationship is created in Data Governance and Catalog to add governance context to the technical asset. - Source System: View assets that are organically related to the technical asset in the catalog source system.
Direction	View related assets of a particular origin. You can select the following values: - Outbound: The relationship is created from the AI model to the related asset. - Inbound: The relationship is created from the related asset to the AI model.
Lifecycle	The lifecycle status of the published asset.
Relationship Type	View assets of a particular direction. You can select the following values: - Direct: View assets that are directly connected to the AI model. - Indirect: View assets that are connected to other assets, which are connected to the AI model that you have opened.

Data Quality tab

View aggregate and individual data quality scores for the data elements that comprise the technical asset.

The following table describes the scores on the **Dimensions Latest Scores** panel:

Dimension	Explanation
Validity	Aggregate score of the data elements with respect to valid and meaningful data.
Accuracy	Aggregate score of the data elements with respect to accurate and useful data.
Completeness	Aggregate score of the data elements with respect to data that is complete and usable.
Timeliness	Aggregate score of the data elements with respect to data that is sufficiently recent for use.
Consistency	Aggregate score of the data elements with respect to data that is consistently and regularly entered.

The following table describes the scores on the **Rules Against Target** panel:

Rules	Explanation
Good	Aggregate scores that are higher than the target value across each dimension for the data elements.
Acceptable	Aggregate scores that are higher than the threshold value but lower than the target value across each dimension for the data elements.
Not Acceptable	Aggregate scores that do not meet the data quality threshold across each dimension for the data elements.

The **Total Rule Occurrence** table shows the individual occurrences of the data quality rule. The aggregate scores on the **Dimensions Latest Scores** and **Rules Against Target** panels are a composite of the individual occurrences in the **Total Rule Occurrence** table. The following table describes the fields that appear in the **Total Rule Occurrence** table:

Field	Description
Name	Name of the rule occurrence.
Dimension	Data quality dimension for which the data quality rule is run.
Latest Score	Aggregate score from the latest rule run.
Date	Date of the latest rule run.
Total Rows	Total number of rows in the data element for which the rule is run.
Invalid Rows	Number of invalid rows in the data element for which the rule is run.
Primary Data Element	Data element on which the data quality rule is run. This data element is the input port for the rule in the integrated data quality system. The data quality score is always generated for the primary data element.
Secondary Data Element	Secondary input port for the data quality rule that is run on the **Primary Data Element**.

Field	Description
Description	Description of the rule occurrence.
Related Rule Template	Rule template for which the rule is run.

Data Observability tab

View the open events for the anomalies identified for technical data sets, data elements, and catalog sources.

The **Open Events by Metric** and **Open Events by Severity** sections represent open events for the data observability anomalies identified in your source data. The identified events are categorized by metrics and severity levels. The **Open Events** table shows all open events for the asset. To view the summary of an event, click the event to open an event page.

The following table describes the fields that appear in the **Open Events** section:

Field	Description
Name	Name of the event that data observability creates when it detects an anomaly in the source data.
Metric	Metric type against which the anomaly was identified and the event was created.
Asset	Source data within which the anomaly was identified. **Note:** The **Asset** field appears for data sets and catalog sources.
Severity	Severity level of the event that data observability identified.
Last Detected	Timestamp that indicates when the anomaly was identified during the catalog source job.
Status	Current state of the event.

Hierarchy tab

The **Hierarchy** tab appears for technical data containers like schemas and databases. You can view the hierarchy of the technical asset in the source system. This tab displays the names of the technical assets in the hierarchy, the type of the technical asset, and the description of the asset.

On the **Hierarchy** tab, the open asset is highlighted.

The page also displays the count of downstream assets for the currently open asset.

Click the expand icon to view the upstream and downstream assets of the technical asset. Click an asset to open the asset on a new page.

Note: In Metadata Command Center, if you modify the properties of the **Extract Group Elements from Hierarchical Files** field and run the catalog source job again, the asset page doesn't display the hierarchical elements in their proper hierarchy groups. To display the assets in their correct hierarchy after you modify the properties of the **Extract Group Elements from Hierarchical Files** field, purge the catalog source before running the job again.

When you delete or purge a catalog source in Metadata Command Center, or when you delete data from a source, the following conditions apply:

- If an existing catalog source is deleted in Metadata Command Center, the technical assets for the catalog source are removed completely from Data Governance and Catalog. If an existing catalog source is purged in Metadata Command Center, the catalog source appears in the browse and search results in Data Governance and Catalog, but any enrichment made to the assets are lost.
- In Metadata Command Center, you can choose whether you want to retain or delete assets in Data Governance and Catalog when the corresponding objects are deleted from the source. If your administrator has chosen to delete objects that are deleted from the source, then the metadata along with the enrichments and relationships to the deleted objects are permanently deleted from the catalog. If your administrator has chosen to retain the objects that are deleted from the source, then Data Governance and Catalog retains the objects that were previously extracted. All enrichments and relationships to the objects are also retained. The objects that are retained in the catalog but deleted from the source appear as unresolved objects inData Governance and Catalog. For more information about configuring the metadata change option while configuring the catalog source, see the *Catalog Source Configuration* help in Metadata Command Center.

Reference Data tab

View and associate reference data from Reference 360 with data elements.

If you have the **View Reference Data** privilege for the Data Governance and Catalog service in Administrator, you can link the asset to a glossary which is associated with reference data or select new reference data from Reference 360. The curated reference data from Reference 360 enhances your business assets and improves data accuracy.

The **Reference Data** tab shows the first 50 code values. To see the complete list of code values, navigate to Reference 360.

If multiple glossaries are associated with reference data, select which glossary to link. To select a different reference data that is more suitable for the data element, click **Switch Reference Dataset**. To remove an already associated reference data, click the **Remove associated reference data** icon. Additionally, you can also view the general properties of the reference data.

Stakeholders tab

The **Stakeholders** tab of a technical asset displays the stakeholder assignment. Stakeholders are authorized users who are responsible for the asset, and can approve or reject change requests for the asset. To view the stakeholders for a technical asset, the organization administrator must provide your user role the **Read** permission on the following Administrator service asset types:

- Folder
- Group
- Organization
- Privilege
- Role
- User

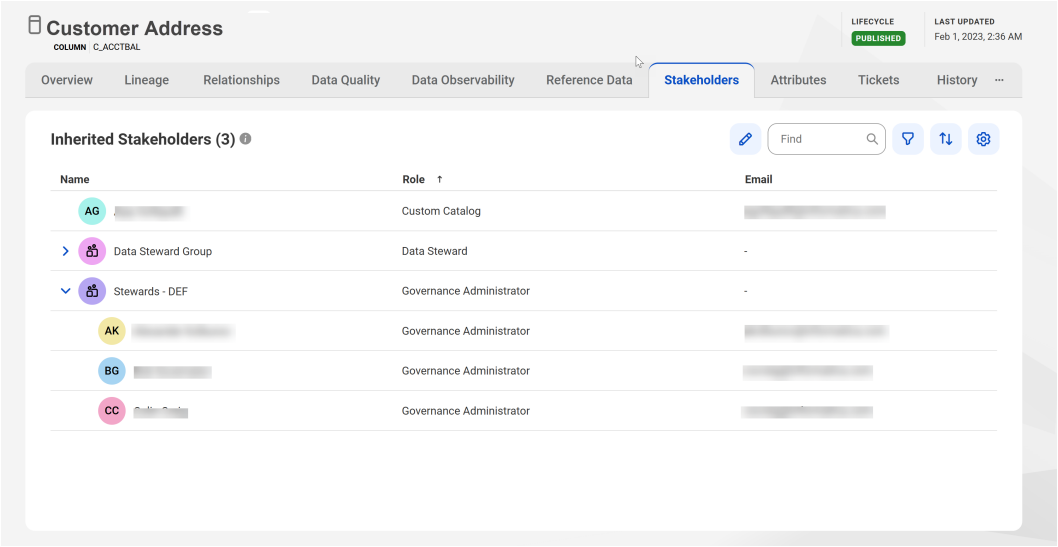
The following table describes the fields that appear on the **Stakeholders** tab of technical assets:

Field	Description
Name	Name of the stakeholder.
Role	The role that is assigned to the stakeholder.
User Origin	The origin of the users. For example, when a user logs in using SSO, the value for this field is displayed as SAML.
Email	The email ID of the user.

The following conditions apply when you assign stakeholders to technical assets:

- If you add or remove stakeholders to an asset, all the existing inherited as well as the newly added stakeholders are displayed as **Assigned** stakeholders. If you add or delete a stakeholder of a child asset, the stakeholder assignment of its parent asset remains unchanged.
- If you delete all inherited stakeholders of an asset, then the asset inherits the stakeholders of the parent asset.
- When you search for `assets with stakeholders`, the search returns the parent assets along with the child assets that have inherited the stakeholders of the parent asset.
- When a user group is assigned as a stakeholder for the parent or child asset, you can view all the stakeholders in the user group by expanding the user group on the **Stakeholders** tab of the asset page.

The following image shows the stakeholder assignment for users and user groups:



Tickets tab

View change request tickets that users have created for the technical asset.

The following table describes the properties of the **Tickets** tab:

Field	Description
Name	Name of the ticket.
Description	Description of the ticket.
Type	Type of ticket.
Status	Status of the change ticket.
Current Task Owner	The stakeholders who are assigned to the tasks in the ticket.

You can filter the list of tickets by selecting available filters options. For more information about tickets, see the *Workflows and tickets* topic in the *Asset Management* help.

History tab

View changes that users have made to the technical asset. The **History** tab provides an audit log that you can use for compliance requirements.

The following table describes the fields that appear on the **History** tab:

Field	Description
Date	Created and last updated dates of the asset.
Event Type	Type of event that is performed on the asset. Event Types include the following: - Asset. Displays changes in the name, description, certification, or the reference ID of the asset. - Relationship. Displays additions, deletions, and the updates of the relationships between the assets. - Stakeholder. Displays additions and deletions of stakeholders to the asset.
Action	Actions performed on the asset: Actions on assets include the following: - Create. Displayed if the asset is created. - Updated. Displayed if the asset is updated. - Deleted. Displayed if the asset is deleted.
Current Attributes	Recently changed attributes of the asset.
Previous Attributes	Earlier attributes of the asset.
Changed By	User who performed any action on the asset.
Asset Name	Name of the asset.
Asset Path	Origin path of the asset.
Asset Type	Type of the asset.

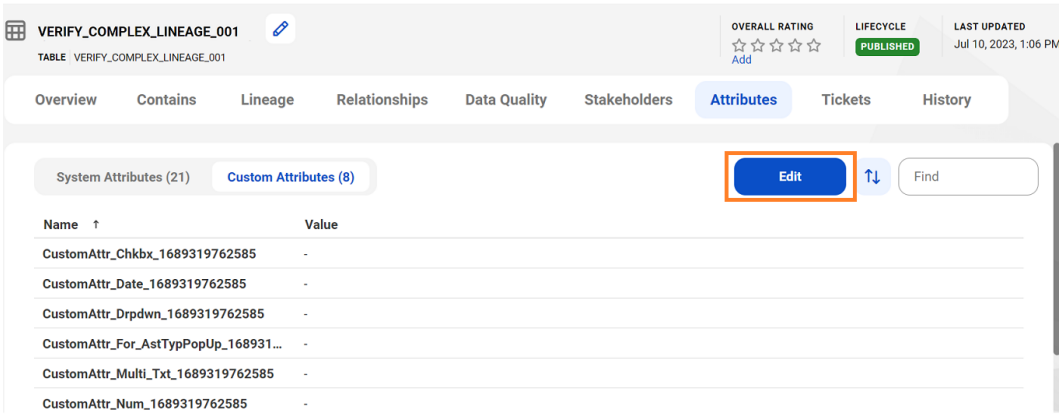
Field	Description
From Asset Type	Source asset type for which the change is applicable. This is applicable when the event type is Relationship.
To Asset Type	Target asset type for which the change is applicable. This is applicable when the event type is Relationship.

Attributes tab

View the system attributes and user custom attributes for technical assets. The system attributes are the predefined properties of the asset. The custom attributes are user-configured properties of the asset that allow you to specify and store specific information relevant to your assets beyond the predefined attributes.

In the **Custom Attributes** sub-tab, you can edit the values of custom attributes .

The following image displays the **Attributes** tab on a technical asset page:



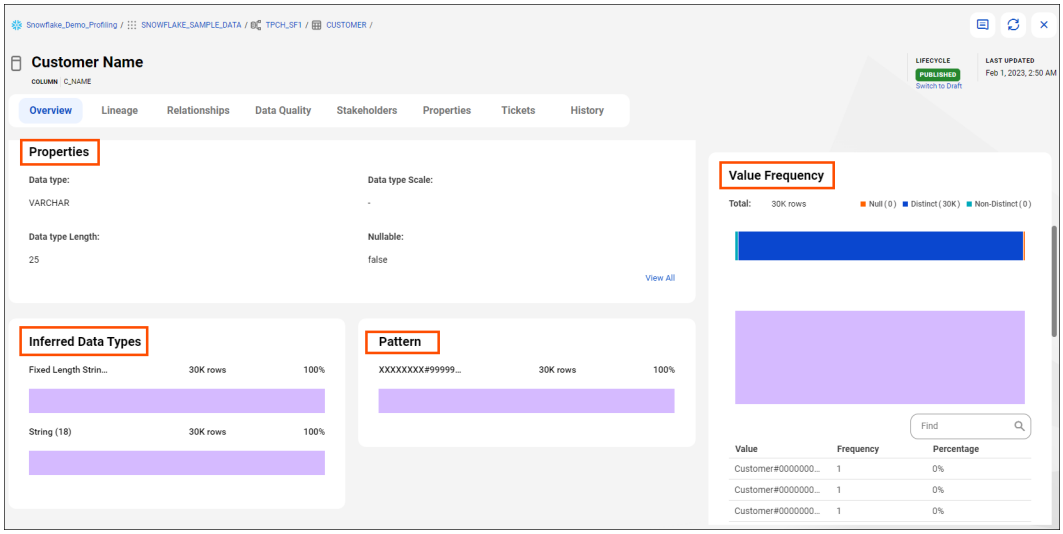
Data profiling statistics

When you enable the data profiling task for a catalog source in Metadata Command Center, the system runs a profile to evaluate the quality of the metadata extracted from the source system. The profile extracts various types of statistics to discover content and structure, such as value distribution, patterns, and data types. The profiling statistics appear in Data Governance and Catalog when you open the technical assets.

You can view profiling statistics and sensitive data if the organization administrator has assigned access policies to you in Metadata Command Center with **Read** permission on the **Profiling** and **Data** attribute groups for the technical assets. For more information about how administrators manage access to assets and attributes, see the *Asset Management* help.

The scope of profiling statistics that Data Governance and Catalog displays depends on the data profiling configuration parameters that you set while configuring a catalog source in Metadata Command Center. If you do not want to run the profile on all the metadata that is extracted from the source system, you can limit the scope of profiling to a subset of metadata by configuring filters for data profiling in Metadata Command Center. For example, you can choose to run data profiling on only three schemas out of the six schemas that the system is configured to extract.

The following sample image shows the data profiling statistics that appear on a column asset page in Data Governance and Catalog after you configure and run the data profiling task for a catalog source in Metadata Command Center:



Properties

This section displays the key data profiling statistics such as the data type of the data element, data type length and scale, and nullable value that indicates if the column can be null or not. You can view more properties of the column on the **Properties** tab.

Inferred data types

This section displays the data types inferred for each data element after you run a profile. It also displays the count of rows that were sampled, along with the percentage of rows containing the data type. Fixed Length String and String are some data types inferred for the column in the above example.

Note: If the precision or scale of a number type value in a column exceeds 28 digits, the precision and scale of the inferred pattern might appear truncated.

Patterns

This section displays the inferred patterns for data elements after you run a profile. Column patterns can include special characters, such as ~, [,], =, -, ?, =, {, *, -, >, <, and $.

Note: If the precision or scale of a number type value in a column exceeds 28 digits, the precision and scale of the inferred data type might appear truncated.

The following table describes the pattern characters and what they represent:

Character	Description
"B" or "b" or " "	Represents a blank space.
"C" or "c"	Represents any character.
"L" or "l"	Represents any lowercase alphabetic character.
{T} or {t}	Represents a tab.
{U} or {u}	Represents any uppercase alphabetic character.

Character	Description
9	Represents any numeric character. Data Governance and Catalog displays up to three characters separately in the "9" format. The tool displays more than three characters as a value within parentheses. For example, the format "9(8)" represents a numeric value with eight digits.
"X" or "x"	Represents any alphabet character. Data Governance and Catalog displays up to three characters separately in the "X" format. The tool displays more than three characters as a value within parentheses. For example, the format "X(6)" might represent the value "Boston." **Note:** The pattern character X is not case-sensitive and might represent uppercase characters or lowercase characters from the source data.
"P" or "p"	Represents "(", the opening parenthesis.
"Q" or "q"	Represents ")", the closing parenthesis.

Value frequency

This section displays the count of Null, Distinct, and Non-distinct values for the data element or the rule after the profile is run. It also displays the frequent values along with the frequency and frequency percentage of each value. Data Governance and Catalog displays the top 20 value frequencies.

Glossary association

To provide user-friendly business context to technical assets, you can associate glossary assets with data elements. A glossary is a collection of business-friendly terms, that when associated with technical assets, helps business users to manage technical data using terminology that is more familiar to them. This leads to consistency and better governance across your organization.

Curate glossaries for technical assets

As a stakeholder, you can curate business glossaries for technical assets to add more business context to your technical metadata and to make your data easily discoverable.

When you curate glossaries, you either accept or decline the glossaries recommended for technical assets, or you manually associate technical assets with one or more glossary assets. Data Governance and Catalog can also automatically associate suitable glossaries with technical assets. You can curate the glossaries on the **Overview** tab or on the **Contains** tab of technical asset pages in Data Governance and Catalog.

The following conditions apply if you want to curate glossaries for technical assets:

- The glossary asset that you curate should be in the **Published** lifecycle stage.
- You can curate glossary recommendations if your organization administrator grants you the **Read** permission on business glossary assets and the **Update** permission on technical assets through access policies in Metadata Command Center.
- The associated glossary asset type can be a business term or a metric.
- The technical asset should be a data element or a technical data set.

Automatically associate glossary assets with technical assets

By using the power of CLAIRE®, Data Governance and Catalog can infer and automatically associate meaningful glossary assets with your technical assets.

Prerequisites

For the system to automatically associate glossary assets with technical assets, your organization administrator must configure the following settings in Metadata Command Center:

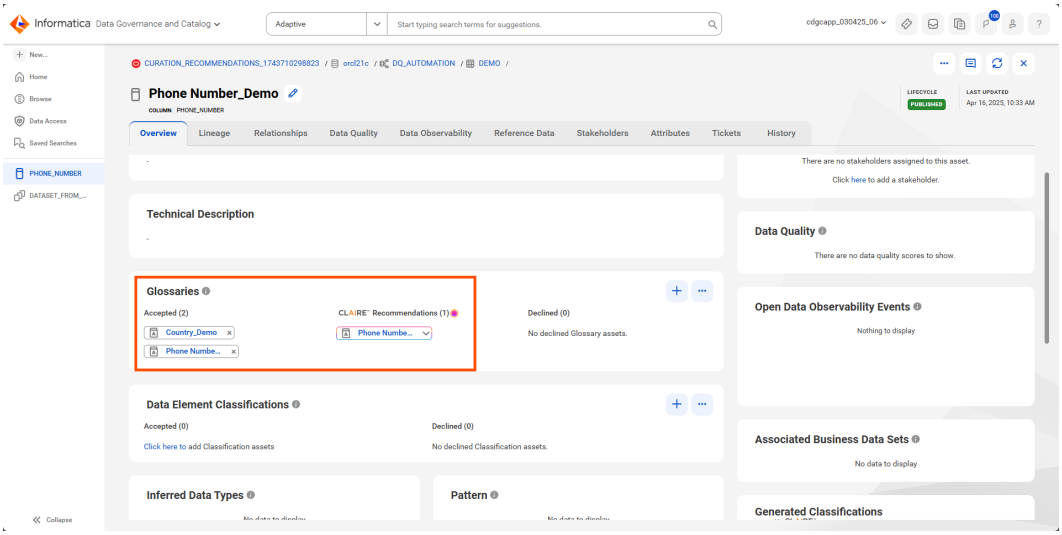
- Enable the **Glossary Association** capability for the catalog source.
- Select the **Enable Auto-Acceptance** option on the **Glossary Association** page.
- Specify the confidence score based on which the glossary assets are automatically accepted for the technical asset.

For information about the glossary association configuration settings, see the *Administration* help in Metadata Command Center.

When you open the **Overview** tab for a technical asset in the catalog source in Data Governance and Catalog, you might see glossary assets in the **Accepted** and **CLAIRE® Recommendations** sections of the Glossaries panel. These associations are made automatically by CLAIRE® based on the following criteria:

- All Glossary assets that match or exceed the specified confidence score are automatically accepted and appear in the **Accepted** list. The accepted glossary asset is thus assigned as the business name of the asset. If there are several automatically accepted glossary assets with the same confidence score, the first asset in the alphabetical order is assigned as the business name of the technical asset.
- All glossary assets that are below the specified confidence score appear as recommendations in the **CLAIRE® Recommendations** list. You can then manually accept or decline the recommendations as mentioned earlier.

The following image shows a sample asset page for a column named `Phone Number_Demo`. The **Glossaries** panel shows the automatically accepted and CLAIRE® recommended terms:

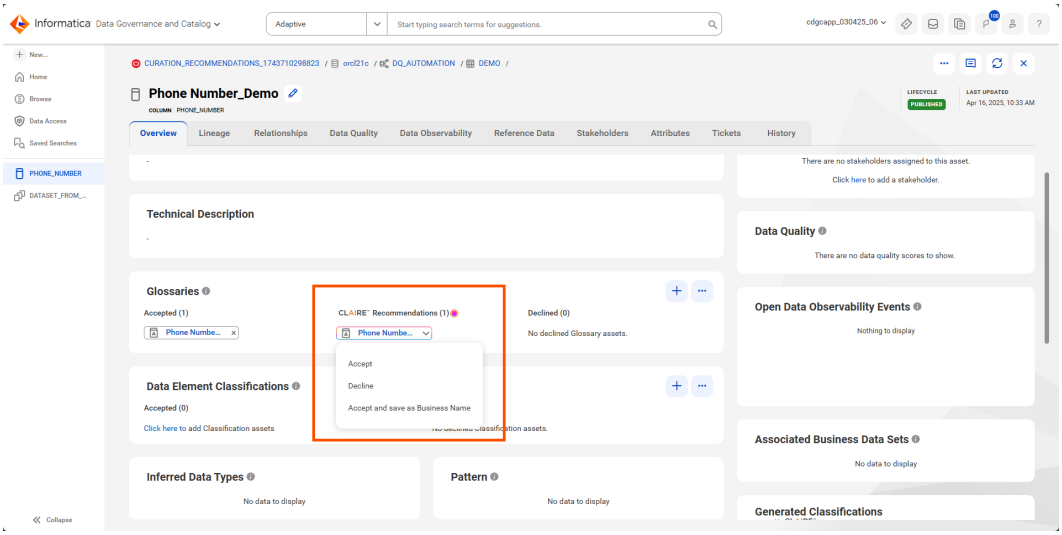


In the above example image, the column named `Phone Number` belongs to an Oracle catalog source. Assume that the confidence score threshold defined for the catalog source in Metadata Command Center is 86%. The business term `Phone Number` matches the column with a confidence score of 88%, and the term is automatically accepted as the business term for the column.

Accept or decline recommended glossary assets

Based on the semantics of data contained in the technical assets, the system can recommend intelligent glossary suggestions for your technical assets. You can either accept or decline the recommendations.

The following image shows the CLAIRE® recommended term and the options for accepting or declining the recommendation for the sample column `PHONE_NUMBER`:

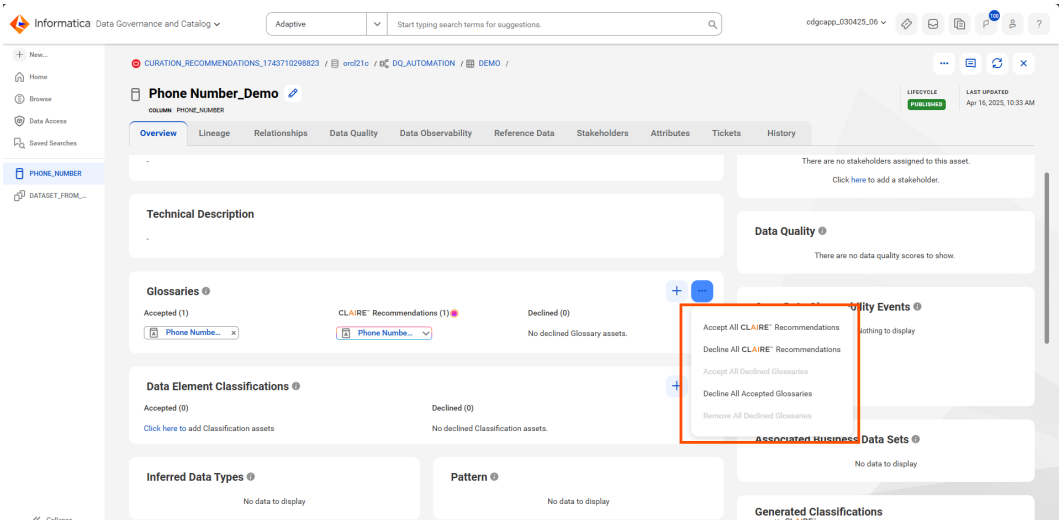


For each CLAIRE® recommendation, you can perform one of the following actions:

- Select **Accept** to accept the CLAIRE® recommendation. The glossary asset then moves to the **Accepted** list.
- Select **Decline** to decline the CLAIRE® recommendation. The glossary asset then moves to the **Declined** list.
- Select **Accept and save as Business Name** to accept the CLAIRE® recommendation and assign this glossary asset as the business name of the asset. The asset then moves to the **Accepted** list and the business name of the asset is assigned.
- Select **Accept and Decline other CLAIRE Recommendations** to accept only this glossary asset and decline other recommended assets. The accepted glossary asset then moves to the **Accepted** list and other recommendations are moved to the **Declined** list.

Optionally, you can bulk accept or decline all CLAIRE® recommended glossary assets at once. You can also bulk accept all the declined glossary assets or decline all accepted glossary assets.

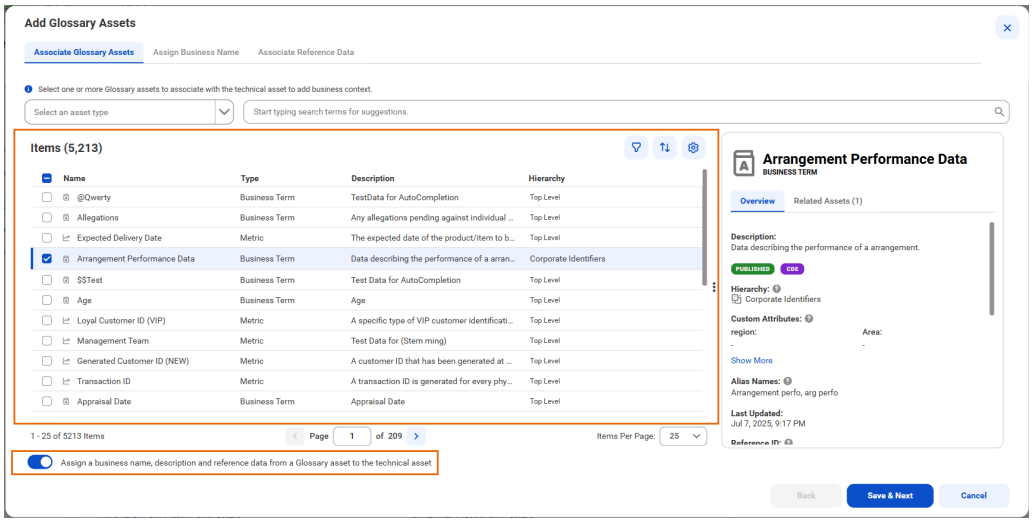
The following image shows the options for bulk accepting and declining glossary assets for a column:



Manually associate glossary assets with technical assets

If you want to associate specific business terms or metrics with technical assets, you can manually associate them in one of the following ways:

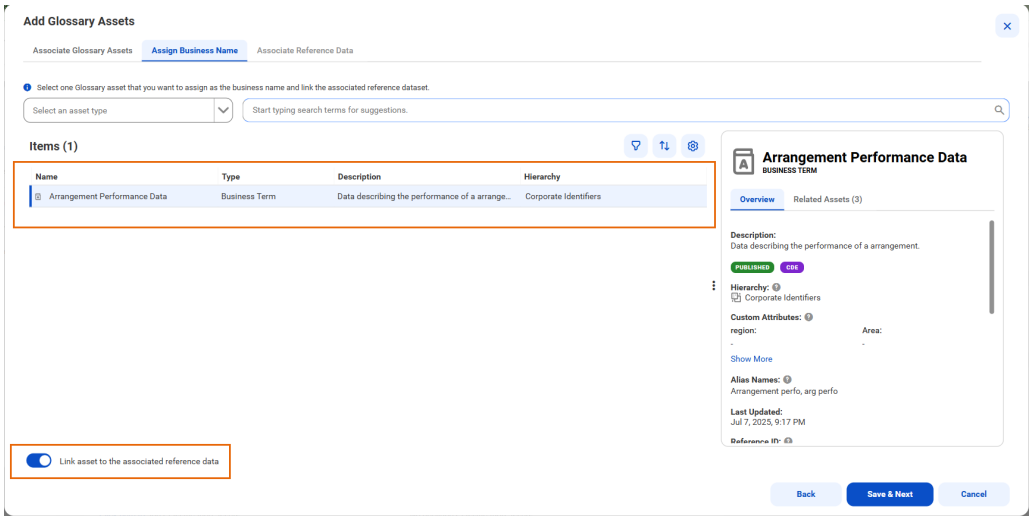
- From the **Overview** tab of a technical asset page. The **Glossaries** panel on the **Overview** tab of a technical asset page displays the glossary assets that are already assigned to the asset. To manually associate other glossary assets, click the add icon on the **Glossaries** panel to open the **Add Glossary Assets** dialog box. In the dialog box, you can select one or more meaningful business terms or metrics to associate with the technical asset and then assign a business name and description for the technical asset.
  The following image shows the **Add Glossary Assets** dialog box that highlights the selected business terms to associate with the technical assets. The dialog box also highlights the option to assign the business name and description of one glossary asset to the technical asset.



After you associate the glossary assets, you can select the option to assign the name and description of one of the shortlisted glossary assets to the technical asset.

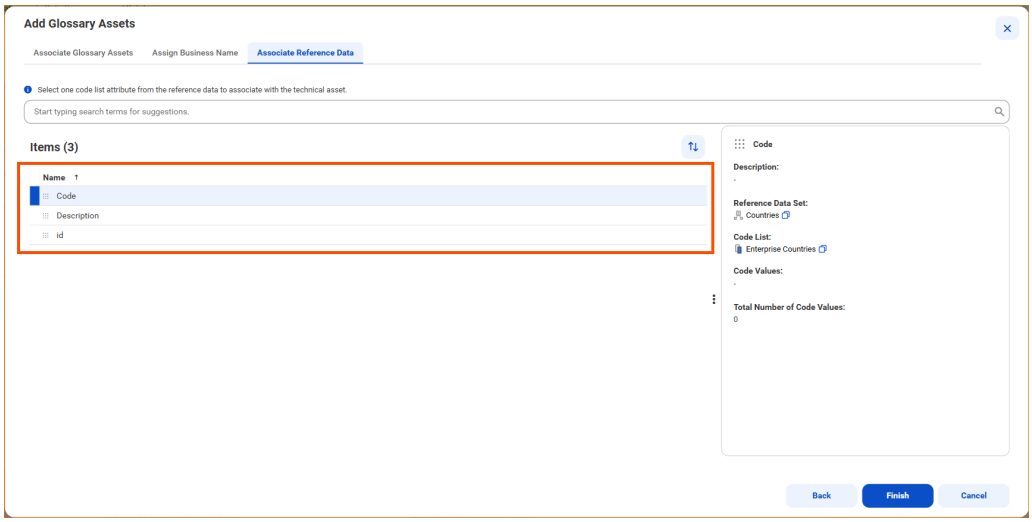
You can then associate the reference data of the selected glossary asset with the technical asset.

The following image shows the **Add Glossary Assets** dialog box with the highlighted option to select one glossary asset to assign as the business name of the technical asset:



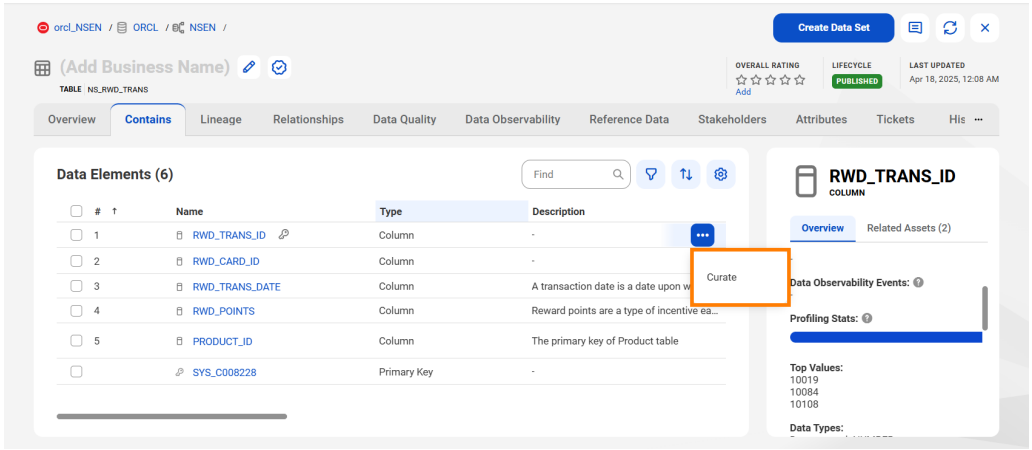
After you assign a business name, you can choose to link the asset to the reference data that is already associated with the glossary.

The following image shows the **Add Glossary Assets** dialog box with the highlighted option to select one code list attribute from the reference data to associate with the technical asset.



After you save and finish the changes, the selected glossary assets appear in the **Accepted** section of the **Glossaries** panel. The business name, description and reference data of the technical asset changes to the name, description and the reference data of the glossary asset that you selected.

- On the **Contains** tab, use the Action menu next to the data element for which you want to curate glossaries to open the **Curate Glossaries and Data Classifications** dialog box. On this dialog box, you can manually associate glossaries with specific data elements.
  The following image shows the **Contains** tab of a table:

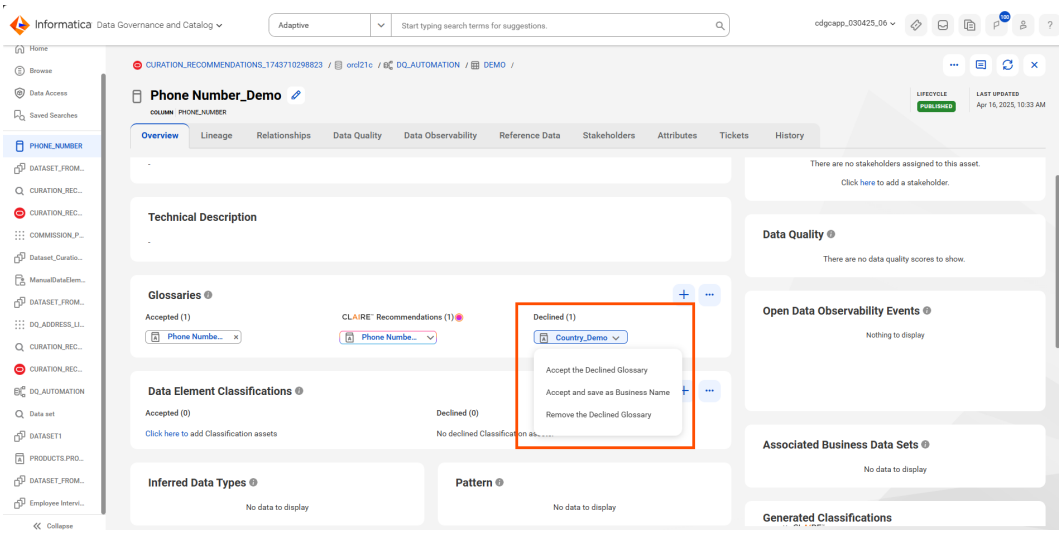


Remove all declined glossary assets

You can remove the glossary assets that are declined for a technical asset.

Optionally, you can bulk remove all of the glossaries. To bulk remove all the declined glossaries for a technical asset, click the action menu and select the appropriate option.

The following image shows the option for bulk removing declined glossary assets:



Data classifications

Data classification is the process of identifying and organizing data into relevant categories based on the functional meaning of data. Classifying data can help your organization manage risks, compliance, and data security.

In Metadata Command Center, enable the data classification capability on a catalog source and create data classification rules to identify and classify data into relevant categories. For example, you can create data classification rules to identify and tag sensitive information such as credit card numbers or customer addresses contained in columns or tables. You can then run the data classification rule against the catalog source and view the classification of columns and tables in Data Governance and Catalog.

For more information about creating data classifications, see the *Administration* help module in Metadata Command Center.

Types of data classifications

Depending on the type of data that you want to identify and classify in your organization, there are various types of classifications that you can create and view in your organization.

Data element classification

This is the smallest unit of data classification. It refers to the classification of columns, fields, structured and unstructured files. Data element classification labels and categorizes information in data elements based on the metadata extracted from source systems and the facts collected as the result of data profiling. For example, you can use data element classification to find sensitive information in assets, such as, credit card numbers, Social Security Numbers or the driver's license numbers. You can then take actions to secure access to sensitive data and set standards for data privacy in your organization.

Data entity classification

A data entity is a collection of data elements and is derived based on an inclusion scope. For example, if "Full Name", "Gender", "Date of Birth", "Email", or "Phone #" are identified in one or more columns of a table, then that table is classified as a 'person' entity. You can use entity classifications to identify important

characteristics of data and group them together as entities. This classification identifies data entities such as purchase order, invoice, customer, location, person, or address contained in a data set.

CLAIRE-generated data classification

This classification is powered by CLAIRE. When you select it, the system automatically generates data classifications for the data elements extracted from a source system. CLAIRE can identify, analyze, and classify the data contained in the data elements without any human input.

When you use rule-based data classifications, you choose from predefined or custom data classifications. When you use CLAIRE-generated data classifications, CLAIRE uses the nomenclature of technical data assets to generate classifications. To generate potential classification labels based on asset names, CLAIRE uses an embedded dictionary.

Generated data classification powered by CLAIRE has the following advantages:

- You can generate data classifications without creating data classification rules. You don't have to know how to create inclusion rules in Metadata Command Center.
- You can generate data classifications even if you don't know which of the existing predefined rule-based data classifications you need to select. CLAIRE automatically generates data classifications for the data elements.
- If you use generated data classification, you don't have to perform profiling. If you use rule-based classifications with Statistics attributes, you need to run data profiling.

For more information about working with generated data classifications, see the *Asset Management* help.

Document classification

This AI-powered classification lets you identify and classify unstructured data into meaningful categories that are specific to your organization. You can configure and run this capability on a catalog source to automatically classify PDFs and text files based on the content of the data.

Document classification has the following advantages:

- Search and discover unstructured data assets alongside existing structured data assets, creating a unified catalog for all your organizational data.
- Fuel your enterprise AI initiatives with well-categorized unstructured data.
- Easily identify sensitive information hidden in documents to meet regulatory compliance standards.

For example, a banking organization stores its loan agreements, mortgage agreements, and risk assessment reports in PDFs and text files in Amazon S3 buckets. After you scan and catalog these documents, you can apply document classification to automatically identify and classify the documents as *Loan Agreement*, *Mortgage Agreement*, and so on based on the structure and content of the documents.

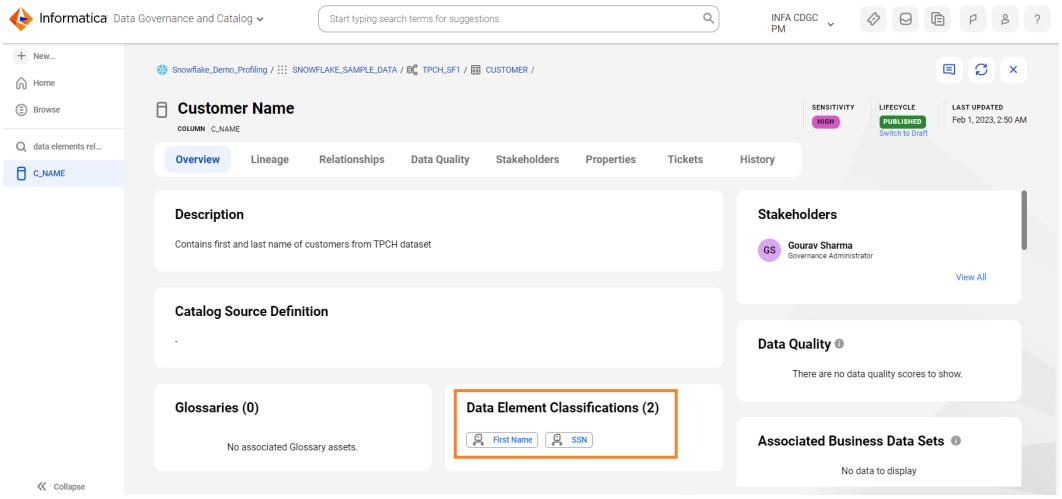
View data classifications for technical assets

You can view data element classifications, data entity classifications, and generated data classifications associated with a technical asset on the asset page.

To view data classifications for technical assets, ensure that the organization administrator grants you the **Read** permission on data classification assets through access policies in Metadata Command Center. For more information about how administrators manage access to assets, see the *Asset Management* help module.

After you run the data classification capability for a catalog source in Metadata Command Center, you can curate the data classifications in Data Governance and Catalog. For more information about curating data classifications for technical assets, see “Curate data classifications for technical assets” on page 28.

The following image shows the data element classifications:



Curate data classifications for technical assets

As a stakeholder, if you want to label or classify data into various categories that are critical to your organization, you can curate data classifications in Data Governance and Catalog.

After you add the data classification rules to a catalog source and run the data classification job for the catalog source in Metadata Command Center, you can curate data classifications for the extracted technical assets in Data Governance and Catalog. When you curate data classifications, you either accept the inferred data classifications to associate them to technical assets, or decline the inferred data classification associations. You can also manually associate technical assets with one or more data classification rules that you have created. For more information about creating data classifications, see the *Administration* help module in Metadata Command Center.

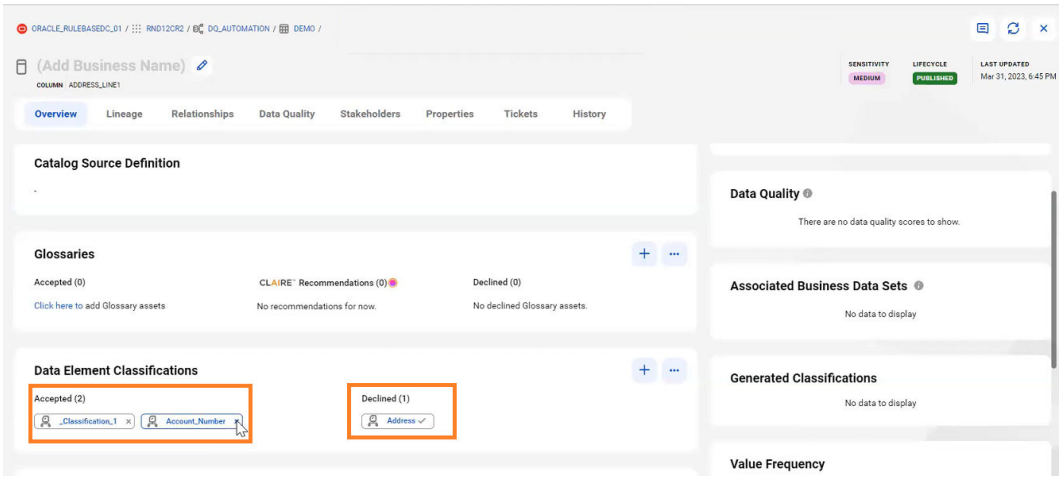
The following guidelines apply if you want to curate data classifications:

- The data classifications that you want to curate should be in the Published **Lifecycle** stage.
- The organization administrator must grant you the **Read** permission on data classification assets and the **Update** permission on technical assets through access policies in Metadata Command Center.
- You can associate data classifications that have been configured with or without inclusion rules in Metadata Command Center.

Accept or decline inferred data classifications

When you add data classification rules to a catalog source in Metadata Command Center, the system identifies the columns and tables that match the rules and displays one or more matched data classifications on the column or table asset pages in Data Governance and Catalog. If your role has the required privileges, you can see the inferred data classifications in the **Accepted** section of the **Data Element Classifications** or **Data Entity Classifications** panel.

The following image shows a column asset page with the inferred data element classifications that match the column data and metadata. All accepted data classification assets appear with an orange color border.



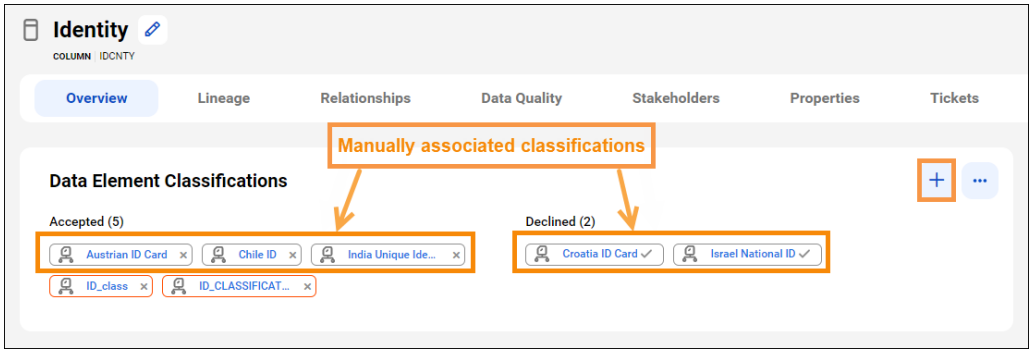
If the data classifications inferred for the columns or tables are not suitable, you can decline the accepted data classifications from the **Accepted** section. The declined classifications move to the **Declined** section of the **Data Element Classifications** or **Data Entity Classifications** panel. The declined classification automatically changes from orange to grey color border. In the above example, three inferred data element classifications were declined for the column.

Manually associate data classifications with technical assets

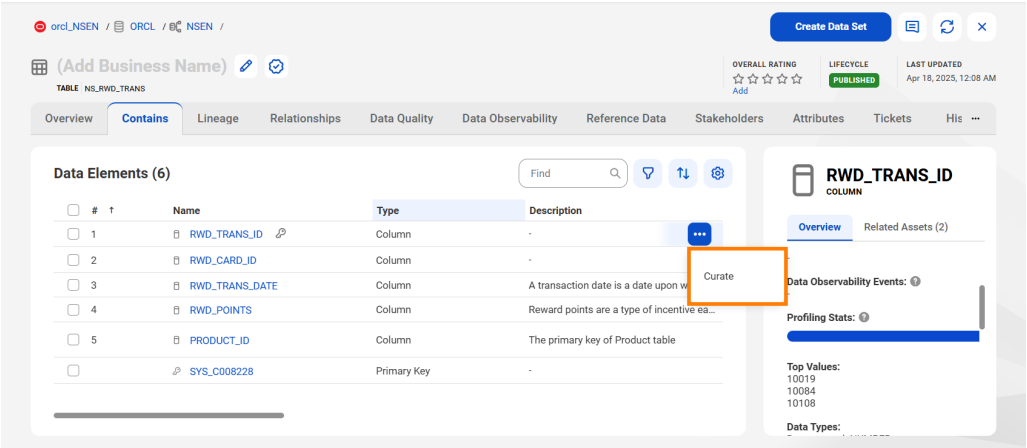
If you did not add data classification rules while configuring the catalog source in Metadata Command Center, you can manually add data classifications to technical assets in Data Governance and Catalog. To manually associate data classifications with technical assets, perform one of the following actions:

- On the **Data Element Classifications** or **Data Entity Classifications** panel, click the add icon, and select one or more data classifications to associate with the data element or the data set.
  The associated data classifications appear in the **Accepted** section of the **Data Element Classifications** panel for a data element or in the **Data Entity Classification** panel for a data set.

  The following image shows the add icon and the manually associated classifications in the **Data Elements Classifications** panel on the column page.



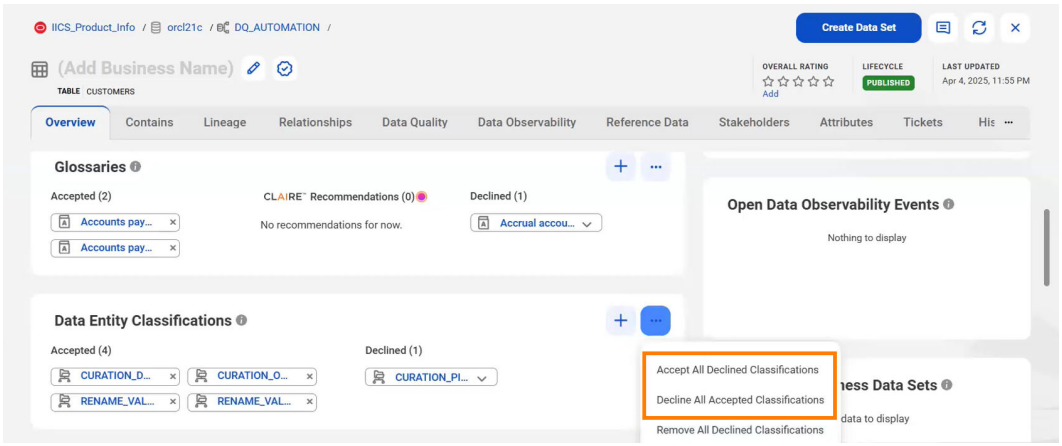
- On the **Contains** tab, use the Action menu next to the data element for which you want to curate data classifications to open the **Curate Glossaries and Data Classifications** dialog box. On this dialog box, you can manually associate data classifications with specific data elements.
  The following image shows the **Contains** tab of a table:



Bulk accept or decline inferred and manually associated data classifications

You can also bulk accept or decline all inferred and manually associated data classifications for a column or a table. To bulk accept or decline data classifications for a technical asset, click the action menu and select the appropriate option.

The following image shows the options for bulk accepting or declining data classifications:

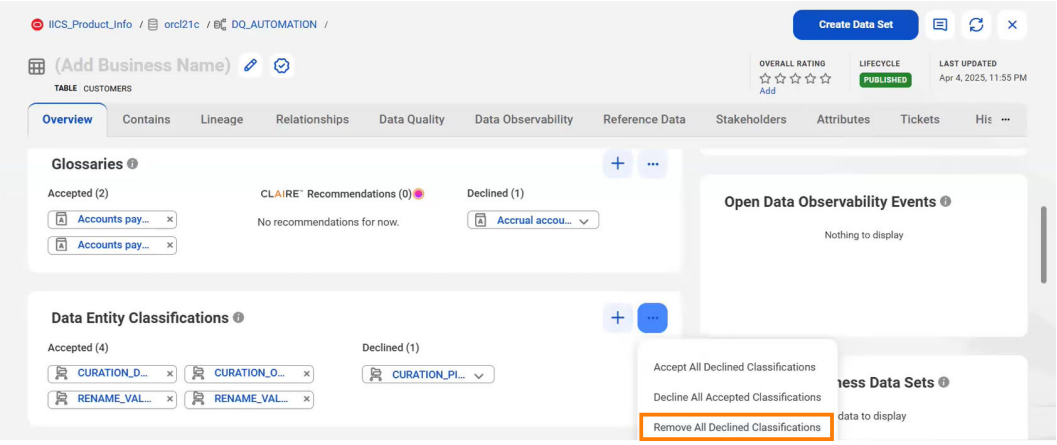


Remove declined data classifications

You can remove the data element classifications and data entity classifications that are declined for a technical asset.

Optionally, you can bulk remove all of the declined data classifications. To bulk remove all the declined classifications for a technical asset, click the action menu and select **Remove All Declined Classifications**.

The following image shows the option for bulk removing declined data classifications:



Curate document classifications for technical assets

As a stakeholder, if you want to label or classify unstructured data into different categories specific to your organization, you can curate document classifications in Data Governance and Catalog. You can use document classification to classify PDFs, text files, and Microsoft Word files.

You can curate document classifications in one of the following ways:

- After you enable and run the document classification job for the catalog source in Metadata Command Center, you can curate document classifications for the extracted technical assets in Data Governance and Catalog. Based on the confidence score threshold configured for the document classifications in Metadata Command Center, the classifications are either automatically accepted or recommended for technical assets in Data Governance and Catalog. When you curate document classifications, you either accept the recommended document classifications to associate them to technical assets, or decline the document classification associations.
- You can also manually associate technical assets with one or more document classifications that you have created. For more information about creating document classifications, see the *Administration* help module in Metadata Command Center.

To curate document classifications, the organization administrator must grant you the **Read** permission on document classification assets and the **Update** permission on technical assets through access policies in Metadata Command Center.

Accept or decline inferred document classifications

When you enable document classification for a catalog source in Metadata Command Center, the system automatically infers the document or file type and classifies them based on their content. If your role has the required privileges, you can see the **Accepted** and **Recommended** document classification on the asset pages in Data Governance and Catalog.

Based on your business context, you can accept or decline recommended document classifications. If the automatically accepted document classifications for the asset are not suitable, you can decline them from the **Accepted** section.

Manually associate document classifications with technical assets

If your administrator has not enabled automatic document classification for the catalog sources in Metadata Command Center, you can manually add document classifications to the assets in Data Governance and Catalog. Your organization administrator creates the document classifications in Metadata Command

Center. In Data Governance and Catalog, you can manually associate the document classifications with assets even if the classification engine is not trained with the document classifications in Metadata Command Center.

To manually associate data classifications with technical assets, go to the **Document Classification** panel of the asset page. Click the add icon, and select one or more document classifications to associate with the asset. The associated classifications appear in the **Accepted** section.

Bulk accept or decline associated document classifications

You can also bulk accept or decline inferred and manually associated document classifications for an asset. Click the action menu and select the appropriate option.

Optionally, you can bulk remove all of the declined data classifications for the technical asset. To bulk remove all the declined classifications for a technical asset, click the action menu and select **Remove All Declined Classifications**.

Data sensitivity levels for assets

Sensitivity labels in Data Governance and Catalog help you classify and protect important data in your organization. This helps an organization understand the value of its data and evaluate whether the data is at risk. It also performs control measures to mitigate risks and helps the organization to comply with relevant industry-specific regulatory mandates.

While creating a data element classification in Metadata Command Center, you can select the sensitivity level to indicate whether an asset is sensitive or non-sensitive. For more information on creating a data element classification, see the *Administration* help in Metadata Command Center.

In Data Governance and Catalog, you can view the sensitivity labels for the data elements that the classification is related to. You can search and find assets based on the sensitivity level by using appropriate search queries. For more information about search query examples, see the *Asset Discovery* help.

View sensitivity of a technical asset

To view the sensitivity of a technical asset in Data Governance and Catalog, the organization administrator must assign the following permission to you through access policies in Metadata Command Center:

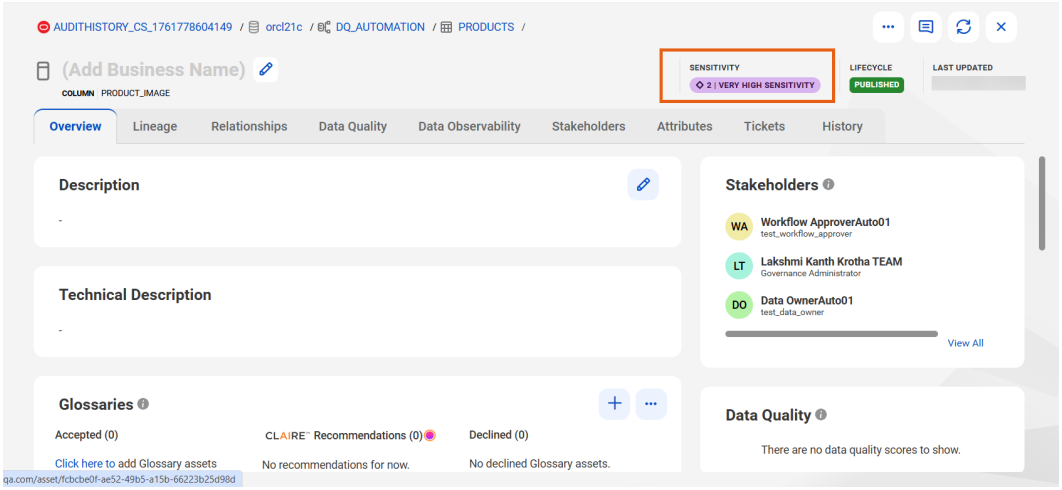
- **Read** permission on technical assets for which you want to view the sensitivity.
- **Read** permission on the data classification assets.

By default, the sensitivity attribute has the following levels:

- **High**. Use for data that is sensitive for your organization. Typically, these can be confidential information, such as credit card details, customer emails and contact numbers, intellectual property, or financial records.
- **Medium**. Use for data that is moderately sensitive for your organization. Typically, these can be internally confidential information, such as employee IDs, technical information, or emails.
- **Low**. Use for data that is not sensitive for your organization. Typically, these can be public information, such as names, designations, qualifications, or public website content. The low sensitivity level is not specifically shown on the Data Governance and Catalog page.

Apart from the predefined sensitivity levels that are listed in this help, you might see other sensitivity levels that your organization administrator can customize exclusively for your organization in Metadata Command Center. For information about how your administrator can customize sensitivity levels, see the *Administration* help in Metadata Command Center.

You can view the sensitivity for data elements on various pages in Data Governance and Catalog. The following image displays the sensitivity of an asset on the asset page:



Data classification categories for assets

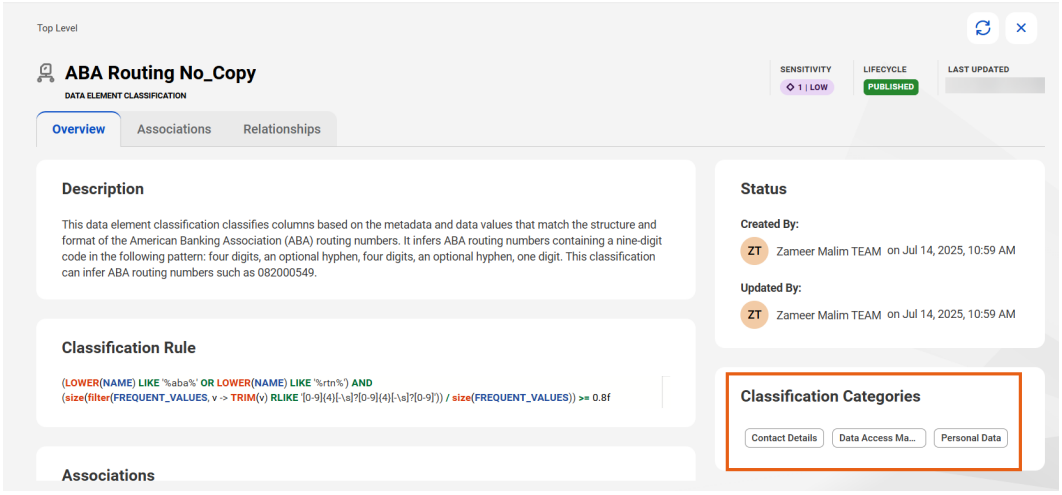
Classification categories put data element classifications into relevant groups. You can add multiple categories for a data element classification in Metadata Command Center. Adding categories to a data element classification helps you identify and curate assets for various business domains.

In Metadata Command Center, you can customize categories for data element classifications. When you create or update a data element classification, you can add one or more categories that are specific to a business domain. For more information on customizing and adding classification categories for data element classifications, see the *Administration* help in Metadata Command Center.

Data Governance and Catalog displays the classification category of data element classifications on various pages. On the data element classification page, the category appears as an attribute of the data element classification. The **Overview** tab for an asset also displays the categories that the data element classifications are associated with. For search-based widgets of the list type, you can select classification categories as a column to display on the widget.

You can search and find assets based on the asset's categories by using appropriate search queries. While searching for assets, you can apply classification categories as a filter to refine your search. For more information about widgets and search see the *Asset Discovery* help.

The following image displays the **Classification Categories** panel on a data element classification page:



CHAPTER 2

Technical asset types

Data Governance and Catalog displays several types of technical assets that are retrieved by Metadata Command Center.

Amazon Athena

Amazon Athena is an interactive query service to query and analyze big data in Amazon S3 using standard SQL.

Objects extracted

Metadata Command Center extracts the following objects from an Amazon Athena source system:

- Database
- Schema
- View
- ViewColumn
- External Table
- External Column
- Calculation
- Resource
- Nested Field

You can extract objects with the following complex data types and their nested fields from an Amazon Athena source system:

- Array
- Map
- Struct

Data Governance and Catalog displays objects with complex data types as external columns. The nested fields within arrays appear as elements, while those within maps appear as keys and values.

Data profiling for Amazon Athena

Configure data profiling to run profiles on the metadata extracted from an Amazon Athena source system. You can run data profiles on the following Amazon Athena objects:

- External tables created in the following file formats:
  - Avro
  - CSV
  - Delta
  - JSON
  - Parquet
- External columns

You can view the profiling statistics in Data Governance and Catalog. The data profiling task runs profiles on the following data types for Amazon Athena objects:

- Bigint
- Boolean
- Char
- Date
- Decimal
- Double
- Float
- Int
- Smallint
- String
- Timestamp
- Tinyint
- Varchar

Sampling type

Determine the sample rows on which you want to run the data profiling task. You can choose one of the following sampling types for an Amazon Athena catalog source:

- All Rows
- Limit N Rows
- Custom Query. Enter the sampling method to specify a percentage of rows on which you want to run the data profiling task. For example, `TABLESAMPLE BERNOULLI(10)` or `TABLESAMPLE SYSTEM(10)`

Note: You can run data quality only on views and external tables that are created in Amazon Athena .

Data lineage

The following lineage data is available for Amazon Athena assets:

- From table to view

For more information about data lineage, see the *Asset Discovery* help.

Amazon Redshift

Amazon Redshift is a cloud-based data warehousing service that you can use to analyze and store data.

Objects extracted

Metadata Command Center extracts the following metadata from an Amazon Redshift source system:

- Database
- Schema
- External Schema
- Table
- External Table
- View
- Materialized View

  Note: Objects of the Materialized View type appear as View in Data Governance and Catalog.
- Function
- Procedure
- Column

Data profiling for Amazon Redshift objects

Configure data profiling to run profiles on the metadata extracted from an Amazon Redshift source system.

You can run data profiles on the following objects:

- Table
- View
- Spectrum External Table

You can run profiles on Spectrum external tables that are created using the following file formats:

- PARQUET
- AVRO
- TEXT
- RC
- SEQUENCE
- ORC

Note: If the Spectrum external tables include columns with string data types, those columns are skipped and are not considered for data profiling and data quality.

The data profiling task runs profiles on the following data types:

- SMALLINT
- INTEGER
- BIGINT
- DECIMAL
- REAL
- DOUBLE PRECISION

- BOOLEAN
- CHAR
- VARCHAR
- DATE
- TIMESTAMP
- STRING
- SUPER

Sampling type

Determine the sample rows on which you want to run the data profiling task. You can choose one of the following sampling types:

- All Rows
- Limit N Rows
- Random N Rows

Data lineage

The following lineage data is available for Amazon Redshift assets:

- From table to view
- Stored Procedures linked to tables or views
- Functions linked to tables or views
- Procedure calling a procedure
- Procedure calling a function

For more information about data lineage, see the *Asset Discovery* help.

Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offered by Amazon Web Services (AWS).

Objects extracted

Metadata Command Center extracts the following metadata from an Amazon S3 source system:

- File
- Folder
- Bucket
- Iceberg Table
- Iceberg Field

You can extract metadata from S3 data sources. By default, you extract metadata from the Amazon S3 Storage, but you can also extract metadata from the S3 Scality Storage or MinIO.

File types

You can extract metadata from the following file types:

- AVRO
- Compressed files
  - TAR
  - ZIP
- Delimited files
  - CSV
  - TSV
- JSON
- Markdown
- Microsoft Excel files
  - Excel 97-2003 Workbook with XLS extension
  - Excel Workbook with XLSX extension
  - Excel Macro-Enabled Workbook with XLSM extension
- Microsoft Word files
  - DOC
  - DOCX
- ORC
- Parquet
- PDF
- TXT
- XML
- XSD

The following table lists the structures associated with the file types that you can extract metadata from:

File Type	Partition structure
AVRO	Single partition, multiple partitions, schema merge
CSV	Single partition, multiple partitions, schema merge
JSON	Single partition, multiple partitions, schema merge
Parquet	Single partition, multiple partitions, schema merge
XML	Single partition, multiple partitions, schema merge

You can extract workbooks, worksheets, and columns from Microsoft Excel files.

You can extract metadata from XML and XSD file formats as XML file objects. Metadata Command Center extracts only elements and attributes from XML and XSD files. If the size of the XML file exceeds 100 KB, Metadata Command Center extracts metadata from the initial 100 KB of the file. However, for XSD file types, Metadata Command Center extracts complete metadata.

Prerequisites for configuring the Amazon S3 catalog source

Use the Amazon S3 v2 connector to connect to the Amazon S3 source system. For information about configuring a connection in Administrator, see *Connections* in the Cloud Common Services help.

Configure permissions

To extract metadata and run profiles, you need account access and permissions on the Amazon S3 source system.

Permissions to extract metadata

Configure the following permissions on the Amazon S3 catalog source:

- Read permission for the user account that you use to access the catalog source.
- Access permission. Required if the user account is different from the user account that you used to create the Amazon S3 catalog source.

Permissions to run data profiles

To perform data profiling, grant the following permissions to the user:

- PutObject. Required to process the staging data for Avro and Parquet files.
- GetObject. Required to read the objects from Amazon S3 buckets.
- DeleteObject. Required to delete the staging data of Avro and Parquet files.
- ListBucket. Required to view the objects from Amazon S3 buckets.
- ListBucketMultipartUploads. Applicable only for mappings in advanced mode.

Data profiling for Amazon S3 objects

Configure data profiling to run profiles on the metadata extracted from an Amazon S3 source system. You can run data profiles on the following Amazon S3 objects:

- AVRO
- CSV
- Parquet

You can view the profiling statistics in Data Governance and Catalog. The data profiling task runs profiles on the following data types for AVRO, CSV, and Parquet file formats:

File format	Data type
Avro	- INT - STRING
CSV	- STRING
Parquet	- BOOLEAN - INT32 - INT64 - FLOAT - DOUBLE - DATE - DECIMAL - STRING

Sampling type

You can run the data profiling task on all rows for an Amazon S3 source system.

Note: To run a profile on an Avro or Parquet file, connect to an advanced cluster. For more information about advanced clusters, see Advanced Clusters help.

Data classification for Amazon S3 objects

Configure data classification for Amazon S3 catalog sources to classify and organize data in your organization. You can view the data classification results in Data Governance and Catalog.

Connection properties

When you configure a connection to the Amazon S3 source system in Administrator, you can view the connection properties for that connection on the **Registration** page in Metadata Command Center.

The following table describes the Amazon S3 connection properties:

Property	Description
Runtime Environment	The execution platform that runs tasks. The runtime environment is either a Secure Agent or a serverless runtime environment.
Access Key	Amazon Web Services access key ID to sign requests that you send to Amazon Web Services.
Secret Key	Amazon Web Services secret access key to sign requests that you send to Amazon Web Services.
Bucket Name	Amazon Web Services bucket name that needs to be scanned.
Region	A named set of Amazon resources in the same geographical area.

Note: To authenticate to Amazon S3 source systems, you can create and use an access key and a secret key. Optionally, you can configure Identity and Access Management (IAM) authentication in your AWS user account. If you use IAM authentication, you do not need to provide access key ID and secret access key explicitly in the Amazon S3 connection. For more information about managing IAM authentication for Amazon S3, see *Amazon S3 Connector* in the Connectors help.

Configuration parameters for metadata extraction

Expand the **Catalog Source Configuration Options** in the **Metadata Extraction** tab of the **Configuration** page. Configure the following parameters for extracting metadata from an Amazon S3 source system:

Parameter	Description
Extract Group Elements from Hierarchical Files	Select one of the following options to extract group or leaf elements from hierarchical files: - Yes. Extracts group elements from hierarchical files with the complete hierarchy of hierarchical fields. You can view the hierarchy of hierarchical files in the **Hierarchy** tab of assets in Data Governance and Catalog. - No. Extracts only leaf elements from hierarchical files without the complete hierarchy of hierarchical fields. You can extract group elements from hierarchical files for the following file types: - AVRO. Extracts and groups hierarchical files and hierarchical fields. - Parquet. Extracts and groups hierarchical files and hierarchical fields. - JSON. Extracts and groups hierarchical files and hierarchical fields. - XML. Extracts and groups elements and attributes. For XML file types, a maximum depth of 1000 elements is permitted within a single element in the hierarchy. - XSD. Extracts and groups elements and attributes. **Attention:** If you modify the **Extract Group Elements from Hierarchical Files** field and run the catalog source again, the asset page doesn't display the hierarchical elements in the correct hierarchy groups. If you modify the property value, purge the catalog source before you run it again.
Enable Extension-Based File Type Detection	Select one of the following options to detect file types by file extensions or by parsing the file contents: - Yes. Detects file types by file extensions. - No. Parses the file contents to detect file types. **Note:** You can detect file types by file extensions for the following file types: - CSV - TSV - TXT - XML
Use First Row as Header of Delimited Files	Select one of the following options to use the first row as the header or detect headers automatically for delimited files: - Yes. Detects column headers based on the following rules: - Duplicate headers get suffixed with '#' followed by a number, for example, ABC#1, ABC#2. The detection is not case-sensitive. - Empty column header values appear as UnknownColumn<position>, for example UnknownColumn2. - The header row in the file is detected even if it has a different number of columns than the data rows. - No. Detects headers automatically for delimited files.
Headers of Delimited Files	Specify values to determine headers of delimited files. Separate multiple values by commas. If any value from the list is found in the first row of the delimited file, then the first row is used as the header. **Note:** This parameter appears only if you choose No for the **Use First Row as Header of Delimited Files** parameter.
Treat Files Without Extension As	Select one of the following options to identify files without an extension: - Parquet - Avro - JSON

Parameter	Description
Enter File Delimiter	Specify the file delimiter if the file from which you extract metadata uses a delimiter other than the following list of delimiters: - Comma(,) - Horizontal tab (\t) - Semicolon (;) - Colon (:) - Pipe symbol (|) Enclose the delimiter in single quotes. For example, '$'. Use a comma to separate multiple delimiters.
Files to be excluded during partition discovery	Specify the regular expression of the files that you want to exclude during partition discovery. Enclose each regular expression in double quotes. For example, ".*json","Customer.csv","Parquet.*". Use a comma to separate multiple regular expressions.

Expand the **Partitioned File Configuration** section in the **Metadata Extraction** tab of the **Configuration** page. The following table shows the parameters that you can configure for custom partition detection and partition pruning:

Parameter	Description
Custom Partition Detection Configuration File	Upload a configuration file in the JSON format to specify the pattern of the non-hive style custom partitions in the source system. Metadata Command Center detects only those partitions that have non-hive style date patterns.
Partitioned Pruning Configuration File	Upload a configuration file in the JSON format. In the file, specify additional information to identify partitions for determining the relevant schema in the source system.
Partition Detection	Select one of the following options: - Yes. Enables partition detection. - No. Disables partition detection.
Partition Pruning	Applicable if you enable partition detection. Select one of the following options: - Yes. Enables partition pruning. - No. Disables partition pruning. Default is Yes.

The following code shows a sample pruned partition configuration file in the JSON format:

```
{
  "SortablePartitionKeys": [
    {
      "Key": "Region",
      "PrunedValues": [
        "USA",
        "IND"
      ]
    }
  ]
}
```

In this example, the partitions `USA` and `IND` are used for determining the schema.

Data lineage

Lineage data is available for Amazon S3 assets that connect to the following source systems:

- Databricks Notebook

- Informatica products, services, and tools
- Microsoft Azure Data Factory
- Microsoft Power BI
- Tableau

For more information about data lineage, see the *Asset Discovery* help.

AWS Glue

Description

AWS Glue is a serverless ETL (extract, transform, and load) service that helps discover, prepare, and integrate data from multiple sources for analysis, machine learning, and application development.

Objects extracted

Metadata Command Center extracts the following metadata from the AWS Glue source system:

- Calculation
- Job
- Job instance

Apache Atlas

Apache Atlas is the governance and metadata framework for Hadoop. Apache Atlas has a scalable and extensible architecture that can be plugged into many Hadoop components to manage their metadata in a central repository.

Objects extracted

Metadata Command Center extracts the following metadata from an Apache Atlas source system:

- Atlas Server
- Hive Process
- Sqoop Process
- Calculation

  Note: Calculation objects are extracted when there is column-level lineage from one asset to another in Hive and Sqoop processes.
- Spark Application
- Spark Process

The Apache Atlas catalog source extracts data lineage from the following data sources:

- Oracle
- MySQL
- PostgreSQL

- Apache Hive
- Hadoop Distributed File System (HDFS)
- Apache HBase

Note: Metadata Command Center skips extraction of Hive processes and the associated lineage links for the following operation types:

- CREATETABLE
- CREATEVIEW
- CREATE_MATERIALIZED_VIEW

Metadata Command Center extracts folders as reference objects from Hadoop Distributed File System.

Metadata Command Center extracts the following objects as reference objects from Apache Hive:

- Schema
- Table
- View
- External Table
- Column

Field and column objects are extracted when there is column-level lineage from one asset to another in Apache Atlas.

Data lineage

The following lineage data is available for Oracle, Hive, and HDFS data sources:

- Data set level
- Data element level

For more information about data lineage, see the *Asset Discovery* help.

Apache Hive

Description

Apache Hive is a data warehouse software built on Apache Hadoop, used to query large datasets from various databases and file systems.

Objects extracted

Metadata Command Center extracts the following metadata from the Apache Hive source system:

- Database
- Schema
- Table
- Column
- External Table
- External Column

- View
- View Column
- Materialized View

  Note: Objects of the Materialized View type appear as View in Data Governance and Catalog.

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Apache HiveQL Script

Apache HiveQL Script is a set of Apache HiveQL statements stored in files that you can use to run sequential scripts.

Objects extracted

Metadata Command Center extracts the following metadata from Apache Hive scripts:

- Calculation
- Folder
- Script
- Statements

Data lineage

The following lineage data is available for Apache HiveQL Script assets:

- Data set level
- Data element level

For more information about data lineage, see the *Asset Discovery* help.

dbt

dbt is a data transformation tool that enables data analysts and engineers to transform data in a cloud data analytics warehouse.

Objects extracted

Metadata Command Center extracts the following metadata from a dbt source system:

- DbtCalculation
- DbtFolder
- DbtModel
  - dbtModelInstance
- DbtPackage
- DbtSeed
- DbtTest
- DbtProject

Note: Metadata Command Center only extracts metadata from models, seeds, and tests.

Data lineage

The following lineage data is available for dbt assets:

- Data set level is available for DbtModel and DbtSeed
- Data element level is available for DbtCalculation

Note: You cannot drill down to view the data element level lineage for the DbtModel and DbtSeed assets. Select a data element and open the **Lineage** tab to view the data element level lineage.

For more information about data lineage, see the *Asset Discovery* help.

Denodo

Denodo is a data virtualization platform that allows access to data from various sources without the need for physical data transfer or replication.

Data sources

Metadata Command Center extracts metadata from Denodo Platform for the following data sources:

- PostgreSQL
- Snowflake
- Oracle
- Microsoft SQL Server
- Microsoft Azure SQL Server
- Databricks
- Google BigQuery

Objects extracted

Metadata Command Center extracts the following metadata from a Denodo source system:

- Database
- Base view
- Derived view

Note: Base views and derived views appear as tables and views in Data Governance and Catalog.

Data lineage

The following lineage data is available for Denodo source system:

- Data set level
- Data element level
- Catalog source level

For more information about data lineage, see the *Asset Discovery* help.

Databricks

Databricks combines data warehouses and data lakes into an AI-driven Databricks Lakehouse platform.

You can run connection-aware scans on Databricks sources.

You can use SQL warehouse or all-purpose clusters to extract metadata.

Note: To improve wildcard lineage at the directory or file level for Databricks assets, perform connection assignment, and run the Databricks catalog source again. These wildcards can refer to files in Amazon S3 and Microsoft Azure Data Lake Storage Gen2.

Objects extracted

You can extract AI Model Core and AI Model Core Version metadata from Databricks Unity Catalog source systems. Additionally, you can retrieve lineage captured by Databricks Unity Catalog.

Note: Databricks Unity Catalog retains lineage data for 90 days.

You can extract metadata from Databricks notebooks if they use the following technologies:

- Markdown
- Python
- SQL

You can extract metadata of the following objects from a Databricks workspace:

- Calculation
- Command
- Experiment
  - Experiment Run
- Folder
- Job Parameter
- Live View
- Materialized View
- Notebook Definition
- Notebook Instance
- Notebook Parameter
- Notebook Task
- Pipeline Definition (SQL based)
  - Live View
  - Materialized View
  - Streaming Table
  - Streaming View
- Result
- Run Job Task
- Streaming Table
- Streaming View
- Task Parameter

- Workflow Job Definition
- Workflow Job Instance

You can extract metadata of the following objects from Databricks Unity Catalog:

- Column
  - Primary Key
  - Foreign Key
- Dashboard
- Database
- External Column
- External Table
- File
- File System
- Function
- Schema
- Table
- View
- Volume

You can extract the following complex data type columns along with their nested fields from Databricks source systems:

- Map
- Struct
- Array

You can extract metadata of the following file types from a Databricks volume:

- PDF
- Text
  - TXT
  - Markdown
- Microsoft Word files
  - DOC
  - DOCX

Data profiling for Databricks objects

Configure data profiling to run profiles on the metadata extracted from Databricks Delta Lake source systems. You can run profiles on Databricks Delta tables created in all-purpose clusters or SQL warehouse. You can also run profiles on Databricks Unity Catalog objects.

You can run profiles on the following Databricks Delta Lake objects:

- Delta Table
- External Table in Delta format.
- View

You can view the profiling statistics in Data Governance and Catalog. The data profiling task runs profiles on the following data types for Databricks Delta Lake objects:

- Bigint
- Boolean
- Date
- Decimal
- Double
- Float
- Int
- Smallint
- String
- Tinyint
- Timestamp

Sampling type

Determine the sample rows on which you want to run the data profiling task. You can choose one of the following sampling types for a Databricks catalog source:

- All Rows
- Custom Query. Sampling method to specify a percentage of rows on which you want to run the data profiling task. For example, **TABLESAMPLE (BUCKET 10 OUT OF 50)**, **TABLESAMPLE (5 PERCENT)**, **LIMIT 2**, or **TABLESAMPLE (5000 ROWS)**.

Data lineage

The following lineage data is available for Databricks assets:

- Data set level
- Data element level

You can extract lineage information from the following source systems:

- Databricks Delta Lake
- Amazon S3
- Oracle
- Microsoft Azure Data Lake Storage Gen2

You can extract lineage information from the following technologies, if available in Unity Catalog:

- Python
- SQL
- Scala
- R

For more information about data lineage, see the *Asset Discovery* help.

erwin Data Modeler File

erwin Data Modeler is a data modeling tool used to find, visualize, design, deploy and standardize data assets.

Objects extracted

Metadata Command Center extracts the following metadata from an erwin Data Modeler source system:

- Argument
- Attribute
- CandidateKeyGroup
- Column
- ColumnAttribute
- Condition
- Db2Database
- DataSource
- DataSourceFile
- DataSourceSchema
- DataSourceTable
- DatabaseSchema
- DefaultValue
- Directory
- Domain
- ERDiagram
- Entity
- File
- FileGroup
- Folder
- ForeignKeyGroup
- Identity
- Index
- IndexMember
- Library
- Mart
- MinMax
- Model
- ModelSource
- NamedRelationshipEnd
- Note
- OracleDatabase
- PrimaryKeyAttribute
- PrimaryKeyColumn

- PrimaryKeyColumnAttribute
- Relationship
- SQLServerDatabase
- Sequence
- StorageGroup
- StoredProcedure
- SubjectArea
- SubtypeRelationship
- SubtypeRelationshipRole
- Synonym
- Table
- TableEntity
- Template
- Trigger
- TypeValue
- UDP
- UDPs
- UnnamedRelationshipEnd,UserDefined
- ValidValuesList
- View
- ViewColumn

erwin Mart Server

erwin Mart Server is a web-based repository to access erwinData Modeler models and perform administrative tasks.

Objects extracted

Metadata Command Center extracts the following metadata from the erwin Mart Server source system:

- Argument
- Attribute
- CandidateKeyGroup
- Column
- ColumnAttribute
- Condition
- DB2Database
- DataSource
- DataSourceFile
- DataSourceSchema

- DataSourceTable
- DatabaseSchema
- DefaultValue
- Directory
- Domain
- ERDiagram
- ModelSource
- NamedRelationshipEnd
- Note
- OracleDatabase
- PrimaryKeyAttribute
- PrimaryKeyColumn
- PrimaryKeyColumnAttribute
- Relationship
- SQLServerDatabase
- Sequence
- StorageGroup
- StoredProcedure
- SubjectArea
- ViewRelationship
- Entity
- File
- FileGroup
- Folder
- ForeignKeyGroup
- Identity
- Index
- IndexMember
- Library
- Mart
- MinMax
- Model
- SubtypeRelationship
- SubtypeRelationshipRole
- Synonym
- Table
- TableEntity
- Template
- Trigger

- TypeValue
- UDP
- UDPs
- UnnamedRelationshipEnd
- UserDefined
- ValidValuesList
- View
- ViewColumn

IDERA ER/Studio Data Architect

IDERA ER/Studio Data Architect is an enterprise tool for data architects to design and document data assets.

Objects extracted

Metadata Command Center extracts the following metadata from the IDERA ER/Studio Data Architect source system:

- Attachment
- Attachment Type
- Attribute
- Candidate Key
- Column
- Condition
- Default
- Diagram
- Domain
- Entity
- File Directory
- Flat File Model
- Flat File Source
- Folder
- Identifying Relationship
- Identity
- Index
- Index Member
- Logical Data Model
- Logical Model
- Logical Source
- Logical Submodel
- Logical Submodel Diagram

- Main Logical Model
- Main Physical Model
- Named Relationship End
- Non Identifying Relationship
- Non Specific Relationship
- Note
- Other Source
- Parameter
- Physical Data Model
- Physical Model
- Physical Source
- Physical Submodel
- Physical Submodel Diagram
- Primary Key Attribute
- Primary Key Column
- Project
- Reference Value List
- Reference Value Range
- Repository
- Rule
- Schema
- Sequence
- Source Table
- Stored Procedure
- Super Type Relationship
- Synonym
- Table
- Transformation
- Trigger
- Type Value
- Unnamed Relationship End
- User
- User Datatype
- View
- View Column
- View Relationship

Note: If you previously extracted objects with no names, you can resolve the issue after you upgrade to the November 2023 release. To resolve the issue, set the **Metadata Change Option** to Delete and then rerun the catalog source job.

File System

Use the File System catalog source to extract metadata from files located on a Windows or a Linux machine into the catalog. You can use either the Local File System protocol to import metadata from the files located on a local machine or use the Remote File System protocol to import metadata from the files located on a remote Windows machine.

Note: To extract metadata from files located on a remote Linux machine, configure the SFTP File System catalog source in Metadata Command Center.

For more information, see the *SFTP File System* catalog source help.

Objects extracted

Metadata Command Center extracts files and folders from a Windows or a Linux file system.

Supported file types

You can extract metadata from the following file types:

- AVRO
- Compressed files
  - TAR
  - ZIP
- Delimited files
  - CSV
  - TSV
- JSON
- Markdown
- Microsoft Excel files
  - Excel 97-2003 Workbook with XLS extension
  - Excel Workbook with XLSX extension
  - Excel Macro-Enabled Workbook with XLSM extension
- Microsoft Word files
  - DOC
  - DOCX
- PDF
- Parquet
- TXT
- XML
- XSD

The following table lists the structures associated with the file types that you can extract metadata from:

File Type	Partition structure
Avro	Single partition, multiple partitions, schema merge
CSV	Single partition, multiple partitions, schema merge
JSON	Single partition
Parquet	Single partition, multiple partitions, schema merge
XML	Single partition, multiple partitions, schema merge

Note: File System catalog sources can only extract input files encoded in UTF-8.

You can extract workbooks, worksheets, and columns from Microsoft Excel files.

You can extract metadata from XML and XSD file formats as XML file objects. Metadata Command Center extracts only elements and attributes from XML and XSD files. If the size of the XML file exceeds 100 KB, Metadata Command Center extracts metadata from the initial 100 KB of the file. However, for XSD file types, Metadata Command Center extracts complete metadata.

Data lineage

Lineage data is available for File System assets that connect to the following source systems:

- Informatica PowerCenter
- Informatica Cloud Data Integration
- Microsoft SQL Server Integration Services (SSIS)

For more information about data lineage, see the *Asset Discovery* help.

Fivetran Platform

Fivetran Platform is a cloud-based data integration platform that automates the extract load transform (ELT) process to simplify analysis and data consolidation from various sources.

Objects extracted

Metadata Command Center extracts the following metadata from a Fivetran Platform source system:

- Connection
- Calculation
- Destination
- Process

Data lineage

The following lineage data is available for Fivetran Platform source system:

- Data set level
- Data element level

- Catalog source level

For more information about data lineage, see the *Asset Discovery* help.

Google BigQuery

Google BigQuery is a serverless data warehouse that performs scalable analysis over petabytes of data. Use a Google BigQuery catalog source to collect metadata from the assets in a Google BigQuery source system.

Objects extracted

Metadata Command Center extracts the following metadata from a Google BigQuery source system:

- Database
- Schema
- Table
- External Table
- Column
- View
- View Column
- Materialized View

  Note: Objects of the Materialized View type appear as View in Data Governance and Catalog.
- Nested Field
- Stored procedure
- SQL User-defined functions (UDF)

You can extract metadata from columns with the following complex data types, along with their nested fields, from Google BigQuery source systems:

- Array
- Range
- Struct

Data Governance and Catalog displays columns with their complex data types and nested fields, including data types of those nested fields. The nested fields within Array and Range column data types appear under element.

Note: Each time Metadata Command Center runs a scan on a Google BigQuery catalog source, Data Governance and Catalog displays the extracted objects along with the objects from the previous scans. This means that if you modify the filters in Metadata Command Center and run a scan, Data Governance and Catalog does not replace the objects from the previous scan. Instead, Data Governance and Catalog adds the newly extracted objects to the existing scanned objects. To see the results of only the latest scan, choose Delete as the Metadata Change Option before you run the scan.

Data profiling for Google BigQuery objects

Configure data profiling to run profiles on the metadata extracted from a Google BigQuery source system. You can run data profiles on the following objects:

- Table

- External Table
- Partitioned Table
- Views
- Materialized Views

The data profiling task runs profiles on the following data types:

- BOOLEAN
- DATE
- DATETIME
- FLOAT
- INTEGER
- NUMERIC
- STRING
- TIME
- TIMESTAMP

Data lineage

The following lineage data is available for Google BigQuery assets:

- From table to view

For more information about data lineage, see the *Asset Discovery* help.

Google BigQuery SQL Script

Google BigQuery SQL Script is a set of Google BigQuery SQL statements stored in files that you can use to run sequential scripts.

Objects extracted

Metadata Command Center extracts the following metadata from Google BigQuery SQL scripts:

- Calculation
- Folder
- Script
- Statement

Data lineage

The following lineage data is available for Google BigQuery SQL Script assets:

- Data set level
- Data element level

For more information about data lineage, see the *Asset Discovery* help.

Google Cloud Storage

Google Cloud Storage is an online file storage web service for storing and accessing data on Google Cloud Platform infrastructure.

Objects extracted

Metadata Command Center extracts the following objects from a Google Cloud Storage source system:

- File
- Project
- Bucket
- Folder
- Hierarchical file (AVRO, JSON, Parquet, XML, XSD)
- Hierarchical field (AVRO, JSON, Parquet)
- Flat file (CSV and Delimited files)
- Flat field (CSV and Delimited files)
- XML File
- XSD File
- Attribute
- Element

Supported file types

You can extract metadata from the following file types:

- AVRO
- CSV
- JSON (Single partition, multiple partitions, schema merge)
- Markdown
- Microsoft Excel files
  - Excel 97-2003 Workbook with XLS extension
  - Excel Workbook with XLSX extension
  - Excel Macro-Enabled Workbook with XLSM extension
- Microsoft Word files
  - DOC
  - DOCX
- Parquet
- PDF
- TSV
- TXT
- XML (XML and XSD files)
- Delimited files
- Flat Partition Detection files

- Multilevel Partition Detection files

The following table lists the structures associated with the file types that you can extract metadata from:

File Type	Partition structure
AVRO	Single partition, multiple partitions, schema merge
CSV	Single partition, multiple partitions, schema merge
JSON	Single partition, multiple partitions, schema merge
Parquet	Single partition, multiple partitions, schema merge
XML	Single partition, multiple partitions, schema merge

You can extract workbooks, worksheets, and columns from Microsoft Excel files.

You can extract metadata from XML and XSD file formats as XML file objects. Metadata Command Center extracts only elements and attributes from XML and XSD files. If the size of the XML file exceeds 100 KB, Metadata Command Center extracts metadata from the initial 100 KB of the file. However, for XSD file types, Metadata Command Center extracts complete metadata.

Data profiling for Google Cloud Storage objects

Configure data profiling to run profiles on the metadata extracted from a Google Cloud Storage source system. You can run data profiles on the following Google Cloud Storage objects:

- AVRO
- CSV
- JSON
- Parquet

You can view the profiling statistics in Data Governance and Catalog. The data profiling task runs profiles on the following data types for AVRO, CSV, JSON, and Parquet file formats:

File format	Supported data type
AVRO	- INT - STRING
CSV	- STRING

File format	Supported data type
JSON	- ARRAY - BIGINT - BOOLEAN - DOUBLE - INTEGER - OBJECT - STRING
Parquet	- BOOLEAN - INT32 - INT64 - FLOAT - DOUBLE - DATE - DECIMAL - STRING

Sampling type

You can run the data profiling task on all rows for a Google Cloud Storage catalog source.

Note: To run a profile on an Avro or Parquet file, connect to an advanced cluster. For more information about advanced clusters, see Advanced Clusters help.

Data classification for Google Cloud Storage objects

Configure data classification for a Google Cloud Storage catalog source to classify and organize data in your organization. You can view the data classification results in Data Governance and Catalog.

Google Looker

Google Looker is a cloud-based enterprise business intelligence tool and a source system from which you can extract metadata about Google Looker business intelligence objects such as looks, and dashboards.

Objects extracted

Metadata Command Center extracts the following metadata from the Google Looker source system:

- Column
- Connection
- ConnectionModel
- Dashboard
- DatabaseSchema
- Dimension
- ExploreDiagram
- Field
- Filter
- Folder

- Join
- Look
- Measure
- Model
- Parameter
- Project
- Query
- QueryField
- Schema
- Server
- Table
- Text
- View

Data lineage

The following lineage data is available for Google Looker Dashboard and Looker assets:

- Data set level
- Data element level

For more information about data lineage, see the *Asset Discovery* help.

Google Vertex AI

Google Vertex AI is a managed platform that you can use to build, train, deploy, and monitor machine learning and generative AI models.

Objects extracted

Metadata Command Center extracts the following metadata from an Google Vertex AI source system:

- Project
- AI Model Core
- AI Model Core Version
- Evaluation Metric

Data lineage

The following lineage data is available for Google Vertex AI assets:

- From dataset to model version

For more information about data lineage, see the *Asset Discovery* help.

Greenplum

Greenplum is a massively parallel processing SQL database based on PostgreSQL open source database technology.

Objects extracted

Metadata Command Center extracts the following metadata from a Greenplum source system:

- Column
  - Primary key
  - Foreign key
- Database
- Schema
- Stored Procedure
- Table
- View
- View Column

Data lineage

The following lineage data is available for Greenplum assets:

- From table to view

For more information about data lineage, see the *Asset Discovery* help.

Hadoop Distributed File System

Hadoop Distributed File System is a distributed file system that handles large data sets that run on commodity hardware.

The Hadoop Distributed File System catalog source is available on the following clusters:

- Cloudera Data Platform (CDP)
- Amazon EMR
- Google Dataproc
- Azure HDInsight

Objects extracted

Hadoop Distributed File System supports the Azure HDInsight, Amazon EMR, Cloudera Data Platform, and Google Dataproc distributions.

Metadata Command Center extracts the following objects from a Hadoop Distributed File System source system:

- File System
- Folder
- File

- Flat File
- Hierarchical File
- Flat Field
- Hierarchical Field
- XML File
- XSD File
- Attribute
- Element

Supported file types

You can extract metadata from the following file types:

- AVRO
- CSV
- JSON
- Markdown
- Microsoft Excel files
  - Excel 97-2003 Workbook with XLS extension
  - Excel Workbook with XLSX extension
  - Excel Macro-Enabled Workbook with XLSM extension
- Microsoft Word files
  - DOC
  - DOCX
- PDF
- Parquet
- TSV
- TXT
- XML (XML and XSD files)

The following table lists the structures associated with the file types that you can extract metadata from:

File Type	Partition structure
AVRO	Single partition, multiple partitions, schema merge
CSV	Single partition, multiple partitions, schema merge
JSON	Single partition, multiple partitions, schema merge
Parquet	Single partition, multiple partitions, schema merge
XML	Single partition, multiple partitions, schema merge

You can extract workbooks, worksheets, and columns from Microsoft Excel files.

You can extract metadata from XML and XSD file formats as XML file objects. Metadata Command Center extracts only elements and attributes from XML and XSD files. If the size of the XML file exceeds 100 KB,

Metadata Command Center extracts metadata from the initial 100 KB of the file. However, for XSD file types, Metadata Command Center extracts complete metadata.

IBM Cognos

Description

IBM Cognos is a business intelligence tool that helps you connect to data sources and create reports.

Objects extracted

Metadata Command Center extracts the following metadata from the IBM Cognos source system:

- ActiveReport
- ActiveReportVariable
- Argument
- Attribute
- BooleanVariable
- CalculatedMeasure
- CalculatedMember
- Calculation
- CalculationShortcut
- ChartNodeMember
- Column
- ConditionFilter
- Connection
- ConnectionPackageDataSource
- CrosstabNodeMember
- Dashboard
- DashboardModel
- DataAttribute
- DataAttributeMeasure
- DataModule
- DataModuleModel
- DataSet
- DataSetModel
- DataSource
- DatabaseConnection
- DatabaseQuery
- DatabaseQueryShortcut
- DatabaseSchema
- DelimitedColumn
- DelimitedFile
- Dimension
- DimensionAttribute
- DimensionAttributeCalculation
- DimensionMap
- DimensionShortcut
- Directory
- DynamicCube
- DynamicSource
- Field
- FieldBody
- Filter
- FilterShortcut
- FixedWidthColumn
- FixedWidthFile
- Folder
- FrameworkManagerModel
- Function
- Hierarchy
- Identifier
- KeyLevelAttribute
- LanguageVariable
- Level
- LevelAttribute
- Macro
- Measure
- MeasureDimension
- MeasureDimensionShortcut
- Member
- Model
- ModelConnection
- ModelQuery
- ModelQueryShortcut
- Namespace
- OlapSchemaNamespace
- Package
- Parameter
- ProcedureQuery
- Prompt
- PromptAnswer
- Query
- QueryItem
- QueryPackage
- QueryStudioModel
- QuerySubject
- Relationship
- Report
- ReportFieldMeasure
- ReportFilter
- ReportFilterCondition
- Repository
- ResultColumn
- ResultSet
- Schema
- ScopeRelationship
- StoredProcedure
- StringVariable
- SummaryStoreConnection
- Table
- Text
- TimeDimension
- Transformer
- TransformerModel
- UserAccount
- View
- ViewColumn

Data Lineage

The following lineage data is available for Report, Table, DatabaseQuery, Query, QueryItem, Column, FieldBody, and DataAttribute assets:

- Data set level
- Data element level

For more information about data lineage, see the *Asset Discovery* help.

IBM InfoSphere DataStage

IBM InfoSphere DataStage is an ETL tool to extract, transform, and load data from multiple sources to target databases or files.

Objects extracted

You can extract metadata from the following DataStage assets:

- Project
- Folder
- Sequence Job
- Parallel Job
- Server Job
- Parallel Shared Container
- Server Shared Container

IBM InfoSphere DataStage jobs consist of stages that you can add to perform specific tasks. Metadata Command Center supports the following stages:

- Parallel job stages:
  - Aggregator
  - Change Capture
  - Checksum
  - Complex Flat File
  - Copy
  - DB2 Connector
  - DRS Connector
  - Data Set
  - FTP
  - FTP Plug In
  - File Connector
  - File Set
  - Filter
  - Funnel
  - Google BigQuery Connector
  - Hierarchical Data

    Note: Partially compatible. You can use it as an internal component, but not as the job input or output.
  - Hive Connector
  - JDBC Connector
  - Join
  - Lookup
  - Merge
  - Modify

- Netezza Connector
- ODBC Connector
- Oracle Connector
- Peek
- Pivot
- Pivot Enterprise
- Remove Duplicates
- Row Generator
- Sequential File
- Slowly Changing Dimensions
- Sort
- Stored Procedure Connector
- Surrogate Key Generator
- Switch
- Teradata Connector
- Transformer
- Unstructured Data
- Xml Output
- Local Container
- Shared Container
- Snowflake Connector

- Server job stages:
  - Aggregator
  - CODBCStage
  - DB2 Connector
  - FTP
  - Hashed File
  - Netezza Connector
  - ODBC Connector
  - Oracle Connector
  - OracleOCI
  - Sequential File
  - Sort
  - Stored Procedure Connector
  - Teradata Connector
  - Transformer
  - Local Container
  - Shared Container

- Sequence job stages:
  - End Loop Activity
  - Execute Command
  - Job Activity
  - Start Loop Activity
  - User Variables

The extracted metadata appear in Data Governance and Catalog as the following assets:

- Project
- Folder
- Job Instance
- Calculation
- Sequence Job
- Parallel Job
- Server Job

Note: Parallel Shared Container and Server Shared Container appear as Calculation under Job Instance in Data Governance and Catalog.

IBM Db2 for LUW

IBM Db2 for LUW is a relational database management system (RDBMS) designed by IBM to store, analyze, and retrieve data from for Windows, Linux and Unix operating systems.

Configure an IBM Db2 for LUW catalog source to extract metadata from an IBM Db2 for LUW database hosted on a dedicated server, or from an Amazon RDS for Db2 database.

Objects extracted

Metadata Command Center extracts the following objects from an IBM Db2 for LUW source system:

- Database
- Schema
- Table
- View
- Column
- Materialized Query Table
- Stored Procedure

Note: To extract the following schemas, you need to define include filters:

- NULLID
- SQLJ
- SYSFUN
- SYSIBMINTERNAL

- SYSIBMTS
- SYSPROC
- SYSPUBLIC
- SYSCAT
- SYSIBM
- SYSIBMADM
- SYSSTAT
- SYSTOOLS

Data lineage

The following lineage data is available for IBM Db2 for LUW assets:

- From table and columns to view

For more information about data lineage, see the *Asset Discovery* help.

IBM Db2 for LUW Script

Metadata Command Center extracts the following metadata from the IBM Db2 for LUW Script source system:

- Calculation
- Folder
- Script
- Statements

Data lineage

The following lineage data is available for Statements and Calculation assets:

- Data set level
- Data element level

Note: You cannot drill down to view the data element level lineage for the Statement assets. Select a data element and open the **Lineage** tab to view the data element level lineage.

For more information about data lineage, see the *Asset Discovery* help.

IBM Db2 for z/OS

IBM Db2 for z/OS is an IBM relational database management system (RDBMS) for IBM Z mainframes to manage core business data. IBM Db2 for z/OS is the main operating system for the IBM hardware platform and IBM zSystems.

Objects extracted

Metadata Command Center extracts the following metadata from the IBM Db2 for z/OS source system:

- Database

- Schema
- Table
- Column
- View
- View column
- Primary key
- Foreign key

Data lineage

Data lineage is available for the following IBM DB2 for z/OS assets at the data set and data element levels:

- Table
- View
- View Column

For more information about data lineage, see the *Asset Discovery* help.

IBM Mainframe Job Control Language (Accelerator)

Job Control Language (JCL) is a scripting language used on IBM mainframe operating systems to instruct the systems on how to run a batch job or start a subsystem. The use of JCL involves the specification of job parameters through the JOB, EXEC, and DD statements to manage input, processing, and output tasks in a structured manner.

Objects extracted

You can use the IBM Mainframe Job Control Language (Accelerator) catalog source to extract metadata from the following objects in the IBM Mainframe Job Control Language source system:

- JCL Job
- JCL Step
- Calculation
- COBOL Program
- COBOL Program Instance
- JCL Procedure
- JCL Procedure Instance
- Partitioned Dataset
- Mainframe File Field
- Mainframe File Format

Data lineage

Data lineage is available for the following IBM Mainframe Job Control Language (Accelerator) assets:

- JCL Step
- COBOL Program Instance
- Mainframe File Field

- Mainframe File Format

For more information about data lineage, see the *Asset Discovery* help.

IBM Netezza

IBM Netezza is a data warehouse system that integrates the database, server, and storage to run and manage complex data warehouse tasks. You can create and configure an IBM Netezza catalog source to extract metadata from an IBM Netezza database.

Objects extracted

Metadata Command Center extracts the following metadata from an IBM Netezza source system:

- Database
- Data Type
- Foreign Key
- Materialized View
- Primary Key
- Schema
- Table
- Unique Key Constraint
- View
- Stored Procedure

Data lineage

The following lineage data is available for IBM Netezza assets:

- From table to view
- From view to view
- From view to table

For more information about data lineage, see the *Asset Discovery* help.

Informatica Developer

Informatica Developer is an application to design and implement data integration, data quality, data profiling, and data services solutions.

Objects extracted

Metadata Command Center extracts the following metadata from an Informatica Developer source system:

- Application
- Calculation
- Catalog Source

- Folder
- Mapping
- Mapping Instance

  Note: Mapping instances include hash codes.
- Mapping Task
- Project
- Workflow

Transformation types

You can extract mappings with the following transformation types:

- Source
- Target
- Lookup
- Sorter
- Normalizer
- Joiner
- Expression
- Rank
- Aggregator
- Sequence
- Union
- Filter
- Router
- Mapplet

Data Lineage

The following lineage data is available for Informatica Developer assets:

- Data set level
- Data element level

For more information about data lineage, see the *Asset Discovery* help.

IDMC

Intelligent Data Management Cloud (IDMC) is an on-demand subscription cloud service that provides access to Cloud Data Ingestion and Replication, Data Integration, MDM SaaS, and Reference 360 assets.

Note:

- For the business entities to appear in the data lineage, create a new Informatica Intelligent Cloud Services catalog source and run the job, instead of running the existing catalog source again.

- When you run an Informatica Intelligent Cloud Services catalog source job, you can view lineage for mappings that read data from and write data to Salesforce source systems.
- You can run an Informatica Intelligent Cloud Services catalog source job to view lineage for mappings that include custom queries to read data from Amazon Redshift and Salesforce source systems.

You can run connection-aware scans on Informatica Intelligent Cloud Services sources.

Objects extracted

Metadata Command Center extracts the following objects from IDMC:

Data Integration

- Advanced mapping
- Application ingestion and replication task
- Application ingestion and replication task instance
- Calculation
- Database ingestion and replication task
- Database ingestion and replication task instance
- Data synchronization task
- Dynamic mapping task
- Dynamic mapping task instance
- Folder
- Mapping
  - ELT (Extract Load Transform) Mapping
  - ETL (Extract Transform Load) Mapping
- Mapping task
- Mapping task instance
- PowerCenter task
- Project

MDM SaaS

- Attribute
- Business entity
  A business Entity can be used in mapping tasks as sources or targets.
- Calculation
- Data Quality
- Data Quality Calculation
- Field
- Folder
- Hierarchy Field
- Hierarchy Instance
- Hierarchy Relationship
- Landing Data Set
- Landing Field

- Mapping
  - ETL (Extract Transform Load) Mapping
- Mapping task
- Mapping task instance
- Match and Survivorship
- Match and Survivorship Calculation
- MDM Hierarchy
- MDM Relationship
- Project
- Relationship Field
- Source Record Data Set
- Source Record Field
- Source System
- Source System Folder
- System field

Reference 360

- Code list
- Crosswalk
- Hierarchy
- Hierarchy data set
- Mapping
  - ETL (Extract Transform Load) Mapping
- Reference data set
- Mapping task
- Mapping task instance

Metadata Command Center extracts the following types of tasks:

- Mapping task. A mapping task defines reusable data flow logic that is not available in Data Synchronization tasks, such as specific ordering of logic or joining sources from different systems.
- PowerCenter task. The PowerCenter task imports PowerCenter workflows to Informatica Cloud and runs them as Informatica Cloud tasks.
- Data synchronization task. The Data Synchronization task synchronizes data between a source and target.
- Mapping task instance. Instance is an execution of a mapping task.

Note: Metadata Command Center also extracts these tasks when they are run from taskflows.

You can extract advanced mappings with the following transformation types:

- Mapplet
- MappletInput
- MappletOutput
- Aggregator
- Sorter

- Expression
- Filter
- Joiner
- Lookup
- Normalizer
- Rank
- Router
- Sequence
- Union
- AccessPolicy

To extract advanced mappings, use the following connectors:

- Amazon Redshift V2
- Amazon S3 V2
- Databricks Delta
- Google BigQuery V2
- Google Cloud Storage V2
- Hive Connector
- Microsoft Azure Data Lake Storage Gen2
- Microsoft Azure Synapse SQL
- Microsoft Fabric Data Warehouse
- Microsoft Fabric Lakehouse
- Microsoft Fabric OneLake
- Snowflake Cloud Data Warehouse

You can override the parameters of a task in a mapping task when you use the following connectors:

- Amazon Redshift V2
- Databricks Delta
- Google BigQuery V2
- Microsoft Azure Synapse SQL
- Microsoft Fabric Data Warehouse
- Microsoft Fabric Lakehouse
- Microsoft Fabric OneLake
- Snowflake Cloud Data Warehouse

To extract metadata from Cloud Data Ingestion and Replication application or database tasks, use the following connectors:

- Application Ingestion and Replication task sources:
  - Salesforce Mass Ingestion

    Note: Use the OAuth 2.0 JWT Bearer Flow authentication.

- Application Ingestion and Replication task targets:
  - Amazon S3
  - Microsoft SQL Server
  - Oracle
  - Snowflake
- Database Ingestion and Replication task sources:
  - Oracle
  - Microsoft SQL Server
- Database Ingestion and Replication task targets:
  - Amazon S3
  - Microsoft SQL Server
  - Oracle
  - Snowflake

Data lineage

The following lineage data is available for IDMC assets (CDI, CDI-E, CDI-R):

- Data set level
- Data element level
- Default column

Data lineage is available for the following MDM SaaS assets when run through Ingress/Egress jobs:

- Data Quality
- Landing Data Set
- Match and Survivorship
- Source Record Data Set

Data lineage is available for the following Reference 360 assets:

- Code list
- Crosswalk
- Hierarchy

For more information about data lineage, see the *Asset Discovery* help.

Informatica PowerCenter

Informatica PowerCenter is an Informatica product that extracts data from multiple sources, transforms the data according to business logic, and loads the transformed data into different target databases.

Objects extracted

Metadata Command Center extracts the following objects from an Informatica PowerCenter source system:

- Folder
- Repository

- Workflow
- Workflow Instance
- Calculation
- Mapping
- Session
- Session Instance

Data lineage

The following lineage data is available for PowerCenter assets:

- Summary lineage
- Data set level
- Data element level
- SQL override
- Pre- and post-SQL session
- Default column

For more information about data lineage, see the *Asset Discovery* help.

JDBC

JDBC is a Java API to connect and interact with databases. You can use a JDBC connection to extract metadata from any source system for which a JDBC driver is available. Metadata Command Center certifies the following source systems from which you can extract metadata using the JDBC catalog source:

- Amazon DynamoDB
- Microsoft Azure Cosmos DB
- Marketo
- Microsoft Dynamics CRM
- MongoDB
- IBM Netezza
- Sybase ASE (Adaptive Server Enterprise)
- Sybase IQ (Intelligent Query)
- Workday
- Apache Cassandra
- SAP SuccessFactors
- NetSuite
- SingleStore
- IBM Db2 for z/OS

To extract metadata from these source systems, download the source-specific JDBC drivers and procure appropriate licenses. For information about using the JDBC drivers and licenses to extract metadata, see the HOW TO: Use the JDBC drivers to extract metadata in Metadata Command Center Knowledge article.

Note: MongoDB and Amazon DynamoDB are NoSQL databases that do not contain views and view columns.

Objects extracted

Metadata Command Center extracts the following metadata from source systems that you connect to with the JDBC catalog source:

- Database
- Schema
- View
- Table
- Column
- View Column

Data lineage

The following lineage data is available for JDBC assets:

- From table to view

For more information about data lineage, see the *Asset Discovery* help.

Kafka

Kafka is a distributed event streaming platform for data pipelines, streaming analytics, data integration, and mission-critical applications.

Configure a Kafka catalog source to extract metadata from Apache Kafka, Confluent Platform, and Confluent Cloud source systems.

Objects extracted

Metadata Command Center extracts the following objects from a Kafka source system:

- Cluster
- Topic
- Field

Metadata Command Center extracts field metadata based on whether a schema registry is configured for the connection. A schema registry centralizes schema storage, management, and version control for data messages in a streaming platform like Kafka.

Metadata Command Center extracts field metadata in the following ways:

- With schema registry: Extracts field metadata from the defined topic schema in Apache Avro format.
- Without schema registry: Extracts field metadata after parsing messages.
  You can extract fields from up to 30 messages per catalog source job in the following data formats:
  - Confluent Cloud: JSON, XML, and CSV
  - Confluent Platform: JSON, XML, and CSV
  - Apache Kafka: JSON, XML, and CSV

Prerequisites for configuring the Kafka catalog source

Use the Kafka connector to connect to Kafka source system. For information about configuring a connection in Administrator, see *Connections* in the Cloud Common Services help.

Connection properties

When you configure a connection to Kafka in Administrator, you can view the connection properties for that connection on the **Registration** page in Metadata Command Center.

On the **Registration** page in Metadata Command Center, choose the connection and choose one of the following Kafka distributions to view the connection properties:

- Apache Kafka. A distributed event streaming platform on premises for data pipelines, streaming analytics, and data integration.
- Confluent Platform. An improved on premises distributed event streaming platform, based on Apache Kafka.
- Confluent Cloud. A fully managed streaming data service on cloud, based on Apache Kafka.

The following table describes the connection properties for the Apache Kafka and Confluent Platform distributions:

Property	Description
Runtime Environment	Name of the runtime environment where you want to run tasks. For more information about how to configure and use the runtime environments, see *Runtime Environments* in the Administrator help.
Kafka Broker List	Comma-separated list of the Kafka brokers. To list a Kafka broker, use the following format: `<HostName>:<PortNumber>` **Note:** When you connect to a Kafka broker over SSL, you must specify the fully qualified domain name for the host name. Otherwise, the test connection fails with SSL handshake error.
Kafka Broker Version	Kafka message broker version. The only valid value is Apache 0.10.1.1 and above.
Additional Connection Properties	Optional. Comma-separated list of additional configuration properties of the Kafka producer or consumer. For a streaming ingestion and replication task, ensure that you set the <kerberos name> property if you configure `<Security Protocol>` as `SASL_PLAINTEXT or SASL_SSL`. For a database ingestion and replication task, if you want to specify a security protocol and properties, specify them here instead of in the **Additional Security Properties** property. For example: `security.protocol=SSL,ssl.truststore.location=/opt/kafka/config/kafka.truststore.jks,ssl.truststore.password=<trustore_password>`.

If you choose the Confluent Cloud distribution, specify values for the following properties on the **Registration** page in Metadata Command Center:

Property	Description
Schema Registry URL	URL to access the schema registry. The URL syntax is `http://host1:port1`.
Confluent Cloud API Key	API key to manage access and authentication to Confluent Cloud.
Confluent Cloud API Secret	API secret for the Confluent Cloud API key.
Confluent Cloud Schema Registry API Key	API key to interact with schema registry in Confluent Cloud.
Confluent Cloud Schema Registry API Secret	API secret for the Confluent Cloud schema registry API key.

Configuration parameters for metadata extraction

Expand the **Catalog Source Configuration Options** in the **Metadata Extraction** tab of the **Configuration** page. Configure the following parameters for extracting metadata from a Kafka source system:

Parameter	Description
Polling Strategy	Select one of the following polling strategies for sampling messages: - From Beginning - From End

MariaDB

MariaDB is an open source Relational Database Management System (RDBMS) solution to store and structure data.

Objects extracted

Metadata Command Center extracts the following metadata from a MariaDB source system:

- Column
- Database
- Schema
- Table
- View
- View Column

Data lineage

The following lineage data is available for MariaDB assets:

- Data set level
- Data element level

For more information about data lineage, see the *Asset Discovery* help.

Microsoft Azure Analysis Services

Microsoft Azure Analysis Services is a managed platform as a service (PaaS) that provides enterprise-grade data models in the cloud.

Objects extracted

Metadata Command Center extracts the following metadata from a Microsoft Azure Analysis Services source system:

- Project
- Table
- Table Calculation
- Measure Group
- Measure
- KPI
- KPI Calculation
- Perspective
- Perspective Table
- Perspective Table Calculation

Data lineage

Data lineage is available for the following Microsoft Azure Analysis Services assets at the data set and data element levels:

- KPI
- KPI calculation
- Measure Group
- Perspective Table
- Perspective Table Calculation
- Table
- Table Calculation

For more information about data lineage, see the *Asset Discovery* help.

Microsoft Azure Blob Storage

Microsoft Azure Blob Storage is a cloud-based object storage for storing unstructured data on Microsoft's data storage platform.

Objects extracted

Metadata Command Center extracts the following metadata from a Microsoft Azure Blob Storage source system:

- Storage Account
- Container

- Folder
- File
- Hierarchical File (AVRO, JSON, Parquet, XML, XSD)
- Flat File (CSV and Delimited files)
- Hierarchical Field (AVRO, JSON, Parquet, XML, XSD)
- Flat Field (CSV and Delimited files)
- Iceberg Table
- Iceberg Field

File types

You can extract metadata from the following file types:

- AVRO
- CSV
- JSON
- Markdown
- Microsoft Excel files
  - Excel 97-2003 Workbook with XLS extension
  - Excel Workbook with XLSX extension
  - Excel Macro-Enabled Workbook with XLSM extension
- Microsoft Word files
  - DOC
  - DOCX
- Parquet
- PDF
- TSV
- TXT
- XML (XML and XSD files)
- Delimited files
- Flat Partition Detection files
- Multilevel Partition Detection files

The following table lists the structures associated with the file types that you can extract metadata from:

File Type	Partition structure
AVRO	Single partition, multiple partitions, schema merge
CSV	Single partition, multiple partitions, schema merge
JSON	Single partition, multiple partitions, schema merge

File Type	Partition structure
Parquet	Single partition, multiple partitions, schema merge
XML	Single partition, multiple partitions, schema merge

You can extract workbooks, worksheets, and columns from Microsoft Excel files.

You can extract metadata from XML and XSD file formats as XML file objects. Metadata Command Center extracts only elements and attributes from XML and XSD files. If the size of the XML file exceeds 100 KB, Metadata Command Center extracts metadata from the initial 100 KB of the file. However, for XSD file types, Metadata Command Center extracts complete metadata.

Microsoft Azure Data Factory

Microsoft Azure Data Factory is a cloud-based ETL and data integration service that is hosted on Microsoft Azure. You can create and configure a Microsoft Azure Data Factory catalog source to extract metadata from the Microsoft Azure Data Factory source system. You can run connection-aware scans on Microsoft Azure Data Factory sources.

Note: To improve wildcard lineage at the directory or file level for Microsoft Azure Data Factory assets, perform connection assignment, and run the Microsoft Azure Data Factory catalog source again. These wildcards can refer to files in Amazon S3 and Microsoft Azure Data Lake Storage Gen2.

Objects extracted

Metadata Command Center extracts the following metadata from a Microsoft Azure Data Factory source system:

- Activity
- Calculation
- Pipeline
- Dataset
- Dataflow
- Resource
- Folder
- Resource Group
- Package
- Data Task
- Factory

Data lineage

The following lineage data is available for Microsoft Azure Data Factory assets:

- Data set level
- Data element level

For more information about data lineage, see the *Asset Discovery* help.

Microsoft Azure Data Lake Storage Gen2

Microsoft Azure Data Lake is a scalable data storage and analytics service that is hosted on Azure. Configure an Azure Data Lake Store (ADLS) catalog source to extract metadata from the data lake store.

Objects extracted

Metadata Command Center extracts the following metadata from an Azure Data Lake Storage Gen2 source system:

- File
- Folder
- Container
- Iceberg Table
- Iceberg Field

File types

You can extract metadata from the following file types:

- AVRO
- Compressed files
  - TAR
  - ZIP
- Delimited files
  - CSV
  - TSV
- JSON
- Markdown
- Microsoft Excel files
  - Excel 97-2003 Workbook with XLS extension
  - Excel Workbook with XLSX extension
  - Excel Macro-Enabled Workbook with XLSM extension
- Microsoft Word files
  - DOC
  - DOCX
- Parquet
- PDF
- TXT
- XML
- XSD

The following table lists the structures associated with the file types that you can extract metadata from:

File Type	Partition structure
Avro	Single partition, multiple partitions, schema merge
CSV	Single partition, multiple partitions, schema merge
JSON	Single partition, multiple partitions, schema merge
Parquet	Single partition, multiple partitions, schema merge
XML	Single partition, multiple partitions, schema merge

You can extract workbooks, worksheets, and columns from Microsoft Excel files.

You can extract metadata from XML and XSD file formats as XML file objects. Metadata Command Center extracts only elements and attributes from XML and XSD files. If the size of the XML file exceeds 100 KB, Metadata Command Center extracts metadata from the initial 100 KB of the file. However, for XSD file types, Metadata Command Center extracts complete metadata.

Data profiling for Microsoft Azure Data Lake Storage Gen2 objects

Configure data profiling to run profiles on the metadata extracted from a Microsoft Azure Data Lake Storage Gen2 source system. You can run data profiles on the following Microsoft Azure Data Lake Storage Gen2 objects:

- Avro
- CSV
- JSON
- Parquet

You can view the profiling statistics in Data Governance and Catalog. The data profiling task runs profiles on the following data types for AVRO, CSV, JSON, and Parquet file formats:

File format	Supported data type
AVRO	- INT - STRING
CSV	- STRING

File format	Supported data type
JSON	- ARRAY - BOOLEAN - NUMBER (double) - NUMBER (float) - NUMBER (int) - NUMBER (long) - OBJECT - STRING
Parquet	- BOOLEAN - INT32 - INT64 - FLOAT - DOUBLE - DATE - DECIMAL - STRING

Sampling type

You can run the data profiling task on all rows for a Microsoft Azure Data Lake Storage Gen2 catalog source.

Note: To run a profile on an Avro or Parquet files, use a secure agent configured with an advanced cluster. The same agent can also process delimited files. If the agent has no advanced cluster capabilities, the data profiling task runs profiles only on delimited files. For more information about advanced clusters, see Advanced Clusters help.

Data classification for Microsoft Azure Data Lake Storage Gen2 objects

Configure data classification for Microsoft Azure Data Lake Storage Gen2 catalog sources to classify and organize data in your organization. You can view the data classification results in Data Governance and Catalog.

Data lineage

Lineage data is available for Microsoft Azure Data Lake Storage Gen2 assets that connect to the following source systems:

- Databricks Notebook
- Informatica products, services, and tools
- Microsoft Azure Data Factory
- Microsoft Power BI
- Snowflake
- Tableau

For more information about data lineage, see the *Asset Discovery* help.

Microsoft Azure SQL Server

Microsoft Azure SQL Server is a managed cloud database.

Objects extracted

Metadata Command Center extracts the following metadata from a Microsoft Azure SQL Server source system:

- Database
- Schema
- Table
- View
- Synonym (local, public)
- Function
- Procedure
- Package
- Trigger
- Column
  - Primary Key
  - Foreign Key
  - Unique
- External Table
- External Column

Data lineage

The following lineage data is available for Microsoft Azure SQL Server assets:

- From table to view
- Stored Procedures linked to tables or views
- Functions linked to tables or views
- Procedure calling a procedure
- Procedure calling a function

For more information about data lineage, see the *Asset Discovery* help.

Microsoft Azure Machine Learning

Microsoft Azure Machine Learning is a cloud-based platform to build, deploy, and manage learning models.

Objects extracted

Metadata Command Center extracts the following metadata from a Microsoft Azure Machine Learning source system:

- AI Model Core
- AI Model Core Version
- Experiment
- Job

- Registry
- Workspace

Data lineage

The following lineage data is available for Microsoft Azure Machine Learning assets:

- Data set level
- Catalog source level

Microsoft Azure Synapse Data Warehouse

Microsoft Azure Synapse Data Warehouse, formerly known as Azure SQL Data Warehouse, is a big data analytic service that queries and analyzes your data. Create a Microsoft Azure Synapse Data Warehouse catalog source to access data from Microsoft Azure SQL Data Warehouse.

Objects extracted

The metadata extraction capability extracts the following objects from a Microsoft Azure Synapse Data Warehouse catalog source:

- Database
- Table
- Schema
- View
- Materialized View
- Function
- Stored Procedure
- Column
- External Table
- External Column

Note: If an incremental metadata extraction job includes a view, it extracts the corresponding tables even if the tables were extracted in previous runs.

Note: Effective in the November 2023 release, Metadata Command Center retrieves the database name from the source system instead of the Azure DW JDBC URL. If you extracted database objects before the November 2023 release, and the name of a database that you entered in the Azure DW JDBC URL uses a different case than the original database name, you might extract duplicate assets when you run the catalog source job again. To ensure that you extract the correct database structure, select **Delete** in the Metadata Change Option and rerun the catalog source job.

Data profiling for Microsoft Azure Synapse Data Warehouse objects

Configure data profiling to run profiles on the metadata extracted from a Microsoft Azure Synapse Data Warehouse source system.

You can run data profiles on the following Microsoft Azure Synapse Data Warehouse objects:

- Tables

- External Tables created in the following file formats on the Microsoft Azure Serverless SQL pool:
  - Delimited text
  - Parquet
- Views

Configure the following data profiling properties in the **Data Profiling** tab of the **Configuration** page in Metadata Command Center:

Property	Description
Azure Blob Container	Name of the Blob container that is used to stage profiled data. Applies to the Azure Blob storage type.
ADLS FileSystem Name	Name of the file system that is used to stage profiled data. Applies to the ADLS Gen2 storage type.
Staging File Format	File format to use when you stage the profiled data. Select one of the following formats: - Delimited Text. Select if the external table is created in the Delimited text file format. - Parquet. Select if the external table is created in the Parquet file format. Applies to external tables created on the Microsoft Azure Serverless SQL pool.
Field Delimiter	Character used to separate fields in a file. Default is `0x1e`. Non-printable characters must be specified as hexadecimal characters. **Note:** Use the field delimiter that you used while creating the external table in the Delimited text file format.
Quote Character	Specifies the quote character to skip when you read data from Microsoft Azure Synapse SQL. The quote character that you specify must not exist in the source table. If it exists, enter a different quote character value. Default is `0x1f`. **Note:** Use the quote character that you used while creating the external table in the Delimited text file format.

You can view the profiling statistics in Data Governance and Catalog. The data profiling task runs profiles on the following data types for Microsoft Azure Synapse Data Warehouse objects:

- Bigint
- Bit
- Char
- Date
- Datetime
- Datetime2
- Decimal
- Float
- Int
- Nchar
- Nvarchar
- Real
- Smalldatetime

- Smallint
- Time
- Tinyint
- Uniqueidentifier
- Varchar

Sampling type

Determine the sample rows on which you want to run the data profiling task. You can choose one of the following sampling types for a Microsoft Azure Synapse Data Warehouse catalog source:

- All Rows
- Limit N Rows
- Random N Rows

Data classification for Microsoft Azure Synapse Data Warehouse objects

Configure data classification for Microsoft Azure Synapse Data Warehouse catalog sources to classify and organize data in your organization. You can view the data classification results in Data Governance and Catalog.

Data Lineage

The following lineage data is available for Microsoft Azure Synapse Data Warehouse assets:

- From table to view
- Stored Procedures linked to tables or views
- Functions linked to tables or views
- Procedure calling a procedure
- Procedure calling a function

For more information about data lineage, see the *Asset Discovery* help.

Microsoft Azure Synapse Analytics

Microsoft Azure Synapse Analytics is an enterprise analytics service that enables insight across data warehouses and big data systems.

Extracted notebook objects

Metadata Command Center extracts the following notebook objects from a Microsoft Azure Synapse Analytics source system:

- Calculation
- Cell
- Folder
- Notebook Definition
- Notebook Instance
- Workspace

You can extract metadata from Microsoft Azure Synapse Analytics notebooks with the following programming languages:

- Python
- SQL

You can extract metadata from Microsoft Azure Synapse Analytics notebooks with the following Python functionalities:

- Standard language constructions
- Standard built-in functions
- Partially-compatible modules:

  Note: Data Governance and Catalog processes only a subset of library functions of partially-compatible modules.

  - abs
  - adal
  - argparse
  - array
  - ast
  - azure
  - base64
  - binascii
  - calendar
  - codecs
  - collections
  - concurrent
  - contextlib
  - contextvars
  - copy
  - copyreg
  - csv
  - dataclasses
  - datetime
  - decimal
  - delta
  - difflib
  - distutils
  - email
  - enum
  - errno
  - fnmatch
  - fractions

- functools
- gc
- genericpath
- gettext
- glob
- graphframes
- hashlib
- heapq
- hmac
- importlib
- inspect
- io
- itertools
- json
- keyword
- locale
- logging
- math
- matplotlib
- mssparkutils
- nt
- numbers
- numpy
- operator
- os
- pandas
- pathlib
- pickle
- pkgutil
- posix
- posixpath
- pprint
- py4j
- pyodbc
- pyspark
- pytz
- random
- re

- reprlib
- requests
- seaborn
- secrets
- shutil
- simplejson
- six
- sklearn
- smtplib
- socket
- ssl
- stat
- string
- struct
- subprocess
- sys
- teradatasql
- textwrap
- threading
- time
- traceback
- types
- typing
- urllib
- urllib3
- uuid
- warnings
- weakref
- xml
- yaml
- zipfile
- zlib

- Custom libraries

  Note: Custom libraries are libraries created by a user.

If a Microsoft Azure Synapse Analytics catalog source detects an incompatible function or library, it can't process the statement. It skips the statement and continues to process the next one.

Extracted pipeline objects

Metadata Command Center extracts the following pipeline objects from a Microsoft Azure Synapse Analytics source system:

- Activity
- Activity Instance
- Calculation
- Folder
- Pipeline
- Pipeline Instance
- Workspace

The Microsoft Azure Synapse Analytics catalog source is compatible with the following Pipeline Activity and Dataset components:

Component name	Description
Copy	Pipeline activity
ExecuteDataFlow	Pipeline activity.
ExecuteSSISPackage	Pipeline activity. Supported page locations: - File System (Package) - Embedded Package
GetMetadata	Pipeline activity
Lookup	Pipeline activity
SqlServerStoredProcedure	Pipeline activity
IfCondition	Pipeline activity
Switch	Pipeline activity
Until	Pipeline activity
ForEach	Pipeline activity
Wait	Pipeline activity
SetVariable	Pipeline activity
ExecutePipeline	Pipeline activity
WebActivity	Pipeline activity
DatabricksNotebook	Pipeline activity
Binary	Dataset

Component name	Description
DelimitedText	Dataset
Excel	Dataset
Json	Dataset
Parquet	Dataset
FileShare	Dataset
AmazonRedshiftTable	Dataset
AzureSqlDWTable	Dataset
AzureSqlTable	Dataset
DynamicsCrmEntity	Dataset
OracleTable	Dataset
SalesforceObject	Dataset
SnowflakeTable	Dataset
SqlServerTable	Dataset
AzureDatabricksDeltaLakeDataset	Dataset

Data lineage

The following lineage data is available for Microsoft Azure Synapse Analytics assets:

- Data set level
- Data element level

For more information about data lineage, see the *Asset Discovery* help.

Microsoft Azure Synapse Script

About catalog source

Microsoft Azure Synapse Data Warehouse Script is a set of Microsoft Azure Synapse SQL statements stored in files that you can use to run sequential scripts.

Objects extracted

Metadata Command Center extracts the following metadata from the Microsoft Azure Synapse Data Warehouse Script source system:

- Calculation
- Folder

- Script
- Statements

Data lineage

The following lineage data is available for Microsoft Azure Synapse Script assets:

- Data set level
- Data element level

For more information about data lineage, see the *Asset Discovery* help.

Microsoft Azure SQL Server Script

Microsoft Azure SQL Server Script is a set of Microsoft Azure SQL statements stored in files that you can use to run sequential scripts.

Objects extracted

Metadata Command Center extracts the following metadata from the Microsoft Azure SQL Server Script source system:

- Calculation
- Folder
- Script
- Statements

Data lineage

The following lineage data is available for Microsoft Azure SQL Server Script assets:

- Data set level
- Data element level

For more information about data lineage, see the *Asset Discovery* help.

Microsoft Fabric Data Warehouse

Microsoft Fabric Data Warehouse is a data warehouse for large-scale enterprise analytics available on the Microsoft Fabric platform.

Objects extracted

Metadata Command Center extracts the following metadata from a Microsoft Fabric Data Warehouse source system:

- Column
- Database
- Function
- Primary Key

- Schema
- Stored procedure
- Table
- View

Data lineage

The following lineage data is available for Microsoft Fabric Data Warehouse assets at the data set level and data element level:

- From table to view
- Stored procedures linked to tables or views
- Functions linked to tables or views
- Procedure calling a procedure
- Procedure calling a function

For more information about data lineage, see the *Asset Discovery* help.

Microsoft Fabric Lakehouse

Microsoft Fabric is a data architecture platform to store, manage, and analyze structured and unstructured data. Use the catalog source to extract metadata from the SQL analytics end point of Lakehouse.

Objects extracted

Metadata Command Center extracts the following metadata from a Microsoft Fabric Data Lakehouse source system:

- Column
- Database
- Function
- Primary Key
- Schema
- Stored procedure
- Table
- View

Microsoft Fabric OneLake

Microsoft Fabric OneLake is a Microsoft Fabric data lake for analytics data.

Objects extracted

Metadata Command Center extracts the following objects from a Microsoft Fabric OneLake source system:

- Workspace
- Lakehouse
- Folder
- File
- Flat File
- Hierarchical File
- Flat Field
- Hierarchical Field
- XML File
- XSD File
- Attribute
- Element

Supported file types

You can extract metadata from the following file types:

- AVRO
- CSV
- JSON
- Markdown
- Microsoft Excel files
  - Excel 97-2003 Workbook with XLS extension
  - Excel Workbook with XLSX extension
  - Excel Macro-Enabled Workbook with XLSM extension
- Microsoft Word files
  - DOC
  - DOCX
- Parquet
- PDF
- TSV
- TXT
- XML (XML and XSD files)

The following table lists the structures associated with the file types that you can extract metadata from:

File Type	Partition structure
AVRO	Single partition, multiple partitions, schema merge
CSV	Single partition, multiple partitions, schema merge

File Type	Partition structure
JSON	Single partition, multiple partitions, schema merge
Parquet	Single partition, multiple partitions, schema merge
XML	Single partition, multiple partitions, schema merge

You can extract workbooks, worksheets, and columns from Microsoft Excel files.

You can extract metadata from XML and XSD file formats as XML file objects. Metadata Command Center extracts only elements and attributes from XML and XSD files. If the size of the XML file exceeds 100 KB, Metadata Command Center extracts metadata from the initial 100 KB of the file. However, for XSD file types, Metadata Command Center extracts complete metadata.

Microsoft OneDrive

Microsoft OneDrive is a file hosting and synchronization service operated by Microsoft as part of its suite of online Office services.

Objects extracted

Metadata Command Center extracts the following objects from a Microsoft OneDrive source system:

- File System
- Folder
- File
- Flat File
- Hierarchical File
- Flat Field
- Hierarchical Field
- XML File
- XSD File
- Attribute
- Element

Supported file types

You can extract metadata from the following file types:

- AVRO
- CSV
- JSON
- Markdown

- Microsoft Excel files
  - Excel 97-2003 Workbook with XLS extension
  - Excel Workbook with XLSX extension
  - Excel Macro-Enabled Workbook with XLSM extension
- Microsoft Word files
  - DOC
  - DOCX
- Parquet
- PDF
- TSV
- TXT
- XML (XML and XSD files)

The following table lists the structures associated with the file types that you can extract metadata from:

File Type	Partition structure
AVRO	Single partition, multiple partitions, schema merge
CSV	Single partition, multiple partitions, schema merge
JSON	Single partition, multiple partitions, schema merge
Parquet	Single partition, multiple partitions, schema merge
XML	Single partition, multiple partitions, schema merge

You can extract workbooks, worksheets, and columns from Microsoft Excel files.

You can extract metadata from XML and XSD file formats as XML file objects. Metadata Command Center extracts only elements and attributes from XML and XSD files. If the size of the XML file exceeds 100 KB, Metadata Command Center extracts metadata from the initial 100 KB of the file. However, for XSD file types, Metadata Command Center extracts complete metadata.

Microsoft Power BI

Microsoft Power BI is a business intelligence tool that connects data and creates reports and dashboards.

Objects extracted

Metadata Command Center extracts the following metadata from a Microsoft Power BI source system:

- Report

  Note: You can also extract metadata from reports generated by Power Query custom functions and DAX functions.
- Workspace
- Dashboard

- Datamart
- Data Set (Semantic Model)
- Tile
- Field
- PaginatedReportDataset
- Dataflow
- Dataflow Table
- Dataset Table

Note: For catalog sources created prior to the November 2024 release, you must run the catalog sources again. When you run the catalog sources again, Metadata Command Center creates a pre-processing job. The pre-processing job updates the extracted metadata to include dataset tables.

Data lineage

Lineage data is available for Microsoft Power BI assets that connect to the following source systems:

- Amazon Redshift
- Amazon S3
- Databricks
- Microsoft Azure SQL Database
- Microsoft Azure Data Lake Storage Gen2
- Microsoft Azure Synapse
- Microsoft SQL Server
- Microsoft SQL Server Analysis Services (SSAS)
- Oracle
- Snowflake
- Teradata Database

The following lineage data is available for Report, Data Set, and Dashboard assets:

- Data set level
- Data element level

The data element level lineage is available for the Field and Title assets.

For more information about data lineage, see the *Asset Discovery* help.

Microsoft SQL Server

Microsoft SQL Server is a Relational Database Management System (RDBMS) solution for storing and retrieving data.

Objects extracted

Metadata Command Center extracts the following metadata from a Microsoft SQL Server source system:

- Database

- Schema
- Table
- View
- Column
  - Primary Key
  - Foreign Key
- Stored Procedure
- Function
- Trigger
- External Table
- External Column
- Objects created using an SQL Server linked server

Data profiling for Microsoft SQL Server objects

Configure data profiling to run profiles on the metadata extracted from a Microsoft SQL Server source system. You can run data profiles on the following Microsoft SQL Server objects:

- Table
- View

You can view the profiling statistics in Data Governance and Catalog. The data profiling task runs profiles on the following data types for Microsoft SQL Server objects:

- BIGINT
- NUMERIC
- BIT
- DECIMAL
- INT
- SMALLINT
- TINYINT
- FLOAT
- REAL
- DATE
- DATETIME2
- SMALLDATETIME
- DATETIME
- TIME
- CHAR
- VARCHAR
- TEXT
- NCHAR
- NVARCHAR
- NTEXT

Sampling type

Determine the sample rows on which you want to run the data profiling task. You can choose one of the following sampling types for a Microsoft SQL Server catalog source:

- All Rows
- Limit N Rows

Data classification for Microsoft SQL Server objects

Configure data classification for Microsoft SQL Server catalog sources to classify and organize data in your organization. You can view the data classification results in Data Governance and Catalog.

Data lineage

The following lineage data is available for Microsoft SQL Server assets:

- From table to view
- Stored Procedures linked to tables or views
- Functions linked to tables or views
- Procedure calling a procedure
- Procedure calling a function

For more information about data lineage, see the *Asset Discovery* help.

Microsoft SQL Server Analysis Services

Microsoft SQL Server Analysis Services is an online analytical processing and data mining tool. It allows you to mine and process data to facilitate decision-making and business analytics.

Objects extracted

Metadata Command Center extracts the following metadata from the SSAS source system:

SSAS model	Extracted assets
Tabular	- Project - Table - Calculation - Measure Group - KPI - Perspective - Perspective Table
Multidimensional	- Project - Data Source View - Data Source View Table - Dimension - Dimension Hierarchy - Cube - Cube Dimension - Cube Dimension Hierarchy - Cube Measure Group - Cube KPI - Cube Partition - Cube Perspective - Cube Calculation

Data lineage

Tabular

The following lineage data is available for Measure Group, KPI, Table, Perspective Table, and Calculation assets:

- Data set level
- Data element level

Multidimensional

The following lineage data is available for Cube Calculation, Cube Dimension, Cube KPI, Cube Measuring Group, Cube Perspective, Data Source View Table, Dimension, Dimension Hierarchy, and Calculation assets:

- Data set level
- Data element level

For more information about data lineage, see the *Asset Discovery* help.

Microsoft SQL Server Analysis Services Multidimensional

Microsoft SQL Server Analysis Services is an online analytical processing and data mining tool. It allows you to mine and process data to facilitate decision-making and business analytics.

Objects extracted

Metadata Command Center extracts the following metadata from the SSAS source system:

SSAS model	Extracted assets
Tabular	- Project - Table - Calculation - Measure Group - KPI - Perspective - Perspective Table
Multidimensional	- Project - Data Source View - Data Source View Table - Dimension - Dimension Hierarchy - Cube - Cube Dimension - Cube Dimension Hierarchy - Cube Measure Group - Cube KPI - Cube Partition - Cube Perspective - Cube Calculation

Data lineage

Tabular

The following lineage data is available for Measure Group, KPI, Table, Perspective Table, and Calculation assets:

- Data set level
- Data element level

Multidimensional

The following lineage data is available for Cube Calculation, Cube Dimension, Cube KPI, Cube Measuring Group, Cube Perspective, Data Source View Table, Dimension, Dimension Hierarchy, and Calculation assets:

- Data set level
- Data element level

For more information about data lineage, see the *Asset Discovery* help.

Microsoft SQL Server Integration Services

Microsoft SQL Server Integration Services (SSIS) is an ETL (Extract, Transform, Load) tool that extracts data from multiple sources, transforms, and loads them into different target databases. SSIS is a component of the Microsoft SQL Server database.

Objects extracted

Metadata Command Center extracts the following objects from a Microsoft SQL Server Integration Services source system:

- Calculation
- Data Task
- Database
- Folder
- Package
- Parameter
- Parameter Container
- Procedure Definition
- Procedure Instance
- Result Set
- Result Set Calculation
- Schema
- Script
- Statement

Data lineage

The following lineage data is available for Microsoft SQL Server Integration Services assets:

- Data set level
- Data element level
- Embedded SQL
- SQL Task
- Stored Procedure calls

For more information about data lineage, see the *Asset Discovery* help.

Microsoft SQL Server Reporting Services

Microsoft SQL Server Reporting Services (SSRS) is a server-based report generating software system. It is part of a suite of Microsoft SQL Server services. You can use SSRS to prepare interactive and printed reports. SQL administrators can connect to SQL databases and use SSRS tools to format SQL reports.

Objects extracted

Metadata Command Center extracts the following objects from a Microsoft SQL Server Reporting Services source system:

- Folders
- Projects
- DataSets

- Fields
- Reports
- PlSql DataSet Statements
- Calculations

Note: If you run catalog sources created prior to the July 2023 release with Retain as the **Metadata Change Option**, duplicate calculations appear in Data Governance and Catalog. To avoid duplicate calculations and to view complete lineage information, run catalog sources created prior to the July 2023 release with Delete as the **Metadata Change Option**.

Data lineage

Lineage data is available for Microsoft SQL Server Reporting Services (SSRS) assets that connect to the following source systems:

- Microsoft SQL Server
- Oracle

For more information about data lineage, see the *Asset Discovery* help.

Microsoft SharePoint online

Microsoft SharePoint Online is a web-based cloud service that integrates with Microsoft Office. It provides a platform to collaborate and allows users to store, share, and manage content.

Objects extracted

Metadata Command Center extracts the following metadata from a Microsoft SharePoint Online source system:

- Site
- Subsite
- Document library
- Folder
- Hierarchical file
- Flat File
- XML File
- XSD File
- Attribute
- Element
- Hierarchical field
- Flat field

File types

You can extract metadata from the following file types using a Microsoft SharePoint Online catalog source:

- AVRO
- CSV

- JSON
- Markdown
- Microsoft Word files
  - DOC
  - DOCX
- Parquet
- PDF
- TSV
- TXT
- XML (XML and XSD files)
- Microsoft Excel files
  - Excel 97-2003 Workbook with XLS extension
  - Excel Workbook with XLSX extension
  - Excel Macro-Enabled Workbook with XLSM extension

The following table lists the structures associated with the file types that you can extract metadata from:

File Type	Partition structure
AVRO	Single partition, multiple partitions, schema merge
CSV	Single partition, multiple partitions, schema merge
JSON	Single partition, multiple partitions, schema merge
Parquet	Single partition, multiple partitions, schema merge
XML	Single partition, multiple partitions, schema merge

You can extract workbooks, worksheets, and columns from Microsoft Excel files.

You can extract metadata from XML and XSD file formats as XML file objects. Metadata Command Center extracts only elements and attributes from XML and XSD files. If the size of the XML file exceeds 100 KB, Metadata Command Center extracts metadata from the initial 100 KB of the file. However, for XSD file types, Metadata Command Center extracts complete metadata.

Microsoft SQL Server Script

Metadata Command Center extracts the following metadata from the Microsoft SQL Server Script source system:

- Calculation
- Folder
- Script
- Statement

Data lineage

The following lineage data is available for Statements and Calculation assets:

- Data set level
- Data element level

Note: You cannot drill down to view the data element level lineage for the Statement assets. Select a data element and open the **Lineage** tab to view the data element level lineage.

For more information about data lineage, see the *Asset Discovery* help.

MicroStrategy

Strategy is an enterprise business intelligence application software and a source system from which you can extract metadata about Strategy business intelligence objects, such as dashboards, scorecards, highly formatted reports.
Strategy was previously known as MicroStrategy. In February 2025, the company including its business intelligence products, was rebranded as Strategy.

Objects extracted

Metadata Command Center extracts the following metadata from the Strategy source system:

- Attribute
- AttributeForm
- CalculatedReportField
- Column
- Consolidation
- ConsolidationElement
- ConsolidationModel
- CustomGroup
- CustomGroupModel
- DatabaseInstance
- DatabaseSchema
- Dataset
- Document
- DocumentReport
- Expression
- Fact
- FactMeasure
- Filter
- FilterModel
- Folder
- HTMLDocument

- HTMLDocumentReport
- HiddenFolder
- Hierarchy
- IntelligentCube
- IntelligentCubeReport
- LogicalQueryTable
- LogicalTable
- Lookup
- Metric
- MetricMeasure
- MetricModel
- Project
- ProjectSource
- Prompt
- PromptAnswer
- PromptModel
- QueryBuilderDefinition
- Report
- ReportMetric
- SQLAttributeForm
- SQLColumn
- SQLMetric
- Schema
- SecurityFilter
- SecurityFilterModel
- Subtotal
- SubtotalMeasure
- SubtotalModel
- Threshold
- WarehouseTable

Data lineage

The following lineage data is available for DocumentReport, HTMLDocumentReport, IntelligentCubeReport, and Report assets:

- Data set level
- Data element level

For more information about data lineage, see the *Asset Discovery* help.

MySQL

MySQL is an open-source relational database management system (RDBMS) to store and structure data.

Objects extracted

Metadata Command Center extracts the following objects from a MySQL source system:

- Database
- Schema
- Table
- View
- Materialized view

Data lineage

The following lineage data is available for MySQL assets:

- From table to view

For more information about data lineage, see the *Asset Discovery* help.

Oracle

Configure an Oracle catalog source to extract metadata from an Oracle database hosted on a dedicated server, an AWS RDS Oracle instance, or Oracle Autonomous Database.
Oracle Autonomous Database is a SaaS implementation of Oracle Database on the Oracle Cloud Infrastructure.

Objects extracted

Metadata Command Center extracts the following objects from an Oracle source system:

- Database
- Schema
- Table
- View
- Materialized View

  Note: Objects of the Materialized View type appear as View in Data Governance and Catalog.
- Column
  - Primary Key
  - Foreign Key
- Stored Procedure
- Function
- Package
- External Table

Note:

- For Oracle On-premise you can extract data from external tables created in File System and Hadoop Distributed File System source systems.
- For Oracle ADB you can extract data from external tables created in Amazon S3, Microsoft Azure Data Lake, and Oracle Cloud Object Storage source systems.

  You can extract data from the external tables created with DBMS_CLOUD.CREATE_EXTERNAL_TABLE and DBMS_CLOUD.CREATE_EXTERNAL_PART_TABLE procedures.

- Sequence
- Trigger
- Objects created using Oracle database link

You can also use the following SQL functionalities:

- Static SQL
- PL/SQL
- Procedure calls with parameters tracing
- Dynamic SQL with metatables
- COPY_DATA
- EXPORT_DATA

Data profiling for Oracle objects

Configure data profiling to run profiles on the metadata extracted from an Oracle source system. You can either use the Oracle connector to establish a connection to Oracle for profiling data, or use the JDBC V2 connector to enable Secure Agent-based data profiling for Oracle. You can run data profiles on the following Oracle objects:

- Table
- View

After running the data profiling capability on an Oracle catalog source, you can view the profiling statistics in Data Governance and Catalog. The data profiling task runs profiles on the following data types for Oracle objects:

- CHAR
- DATE
- NUMBER
- NUMBER (p,s)
- TIMESTAMP
- VARCHAR
- VARCHAR2
- NCHAR
- NVARCHAR2
- FLOAT (126)
- TIMESTAMP WITH TIMEZONE
- TIMESTAMP WITH LOCAL TIMEZONE
- INTERVAL DAY TO SECOND

- INTERVAL YEAR TO MONTH
- BINARY FLOAT
- BINARY DOUBLE
- FLOAT
- UROWID
- ROWID

Note: Oracle supports the maximum precision value of 38 for the NUMBER data type for data profiling.

Sampling type

Determine the sample rows on which you want to run the data profiling task. You can choose one of the following sampling types for an Oracle catalog source:

- All Rows
- Limit N Rows

Data classification for Oracle objects

Configure data classification for an Oracle catalog source to classify and organize data in your organization. You can view the data classification results in Data Governance and Catalog.

Data lineage

The following lineage data is available for Oracle assets:

- From table to view
- Stored Procedures linked to tables or views
- Functions linked to tables or views
- Procedure calling a procedure
- Procedure calling a function

For more information about data lineage, see the *Asset Discovery* help.

Oracle Business Intelligence

Oracle Business Intelligence is a business intelligence tool to gather, store, and visualize enterprise data. You can create and configure an Oracle Business Intelligence catalog source to extract metadata from the Oracle Business Intelligence Enterprise Edition Presentation Server. Metadata Command Center extracts report objects that include presentations, business, and physical models.

Objects extracted

Metadata Command Center extracts the following metadata from an Oracle Business Intelligence source system:

- Alias
- Alias Column
- Alias Key Column
- Alias Table
- Answers Report

- Business Model
- Business Model and Mapping
- Catalog
- Column
- Condition
- Criteria
- Criteria Column
- Criteria Hierarchical Column
- Dashboard
- Dashboard Page
- Dashboard Prompt
- Database
- Data Set
- Data Source
- Derived Logical Column
- Derived Presentation Column
- Element
- External Schema
- Folder
- Group
- Hierarchy
- Key
- KPI
- Link
- List of Values
- Logical Column
- Logical Dimension
- Logical Fact Table
- Logical Key Column
- Logical Level
- Logical Table
- Measure
- Parameter
- Parameter Answer
- Parameter Value
- Physical Column
- Physical Key Column
- Physical Table
- Presentation Column

- Presentation Fact Table
- Presentation Hierarchy
- Presentation Level
- Presentation Table
- Presentation Variable
- Prompt
- Prompt Answer
- Publisher Data Model
- Publisher Report
- Report
- Report Field
- Repository
- Repository Variable
- Session Initialization Block
- Session Variable
- Static Variable
- Subject Area
- Value
- Variable

Data lineage

Data lineage is available for the following Oracle Business Intelligence assets at the data set and data element levels:

- Alias Column
- Alias Key Column
- Alias Table
- Answers Report
- Criteria
- Criteria Column
- Logical Column
- Logical Dimension
- Logical Fact Table
- Logical Key Column
- Logical Table
- Measure
- Physical Key Column
- Physical Table
- Presentation Column
- Presentation Fact Table
- Presentation Hierarchy

- Presentation Level
- Presentation Table
- Presentation Variable
- Presentation Variables
- Prompt Answer
- Report
- Report Field

For more information about data lineage, see the *Asset Discovery* help.

Oracle Cloud Object Storage

Oracle Cloud Object Storage is a cloud data storage service. It comprises several components that work together to provide data-driven insights.

Objects extracted

Metadata Command Center extracts the following metadata from an Oracle Cloud Object Storage source system:

- OCI Bucket
- Folder
- File
- Flat File
- Hierarchical File
- Flat Field
- Hierarchical Field
- XML File
- XSD File
- Attribute
- Element
- Workbook
- Worksheet
- Column

File types

You can extract metadata from the following file types using an Oracle Cloud Object Storage catalog source:

- AVRO
- CSV
- JSON
- Markdown

- Microsoft Excel files
  - Excel 97-2003 Workbook with XLS extension
  - Excel Workbook with XLSX extension
  - Excel Macro-Enabled Workbook with XLSM extension
- Microsoft Word files
  - DOC
  - DOCX
- Parquet
- PDF
- TSV
- TXT
- XML (XML and XSD files)
- ZIP

The following table lists the structures associated with the file types that you can extract metadata from:

File Type	Partition structure
AVRO	Single partition, multiple partitions, schema merge
CSV	Single partition, multiple partitions, schema merge
JSON	Single partition, multiple partitions, schema merge
Parquet	Single partition, multiple partitions, schema merge
XML	Single partition, multiple partitions, schema merge

You can extract workbooks, worksheets, and columns from Microsoft Excel files.

You can extract metadata from XML and XSD file formats as XML file objects. Metadata Command Center extracts only elements and attributes from XML and XSD files. If the size of the XML file exceeds 100 KB, Metadata Command Center extracts metadata from the initial 100 KB of the file. However, for XSD file types, Metadata Command Center extracts complete metadata.

Oracle Golden Gate

Oracle Cloud Infrastructure GoldenGate is a data-replication tool to load, distribute, and filter committed transaction data within an enterprise and to enable migration between databases. You can also perform continuous extraction and replication of transactional Data Manipulation Language (DML) operations and data definition language (DDL) changes to keep your source and target data consistent.

Objects extracted

Metadata Command Center extracts the following metadata from an Oracle Cloud Infrastructure GoldenGate source system:

- GoldenGateJob

- Calculation
- Replication Job
- Replication Process

The following attributes are extracted for Replication Job and Replication Process:

- Host
- Trail file name
- Trail file host

Oracle PL/SQL Script

Oracle PL/SQL Script is a set of Oracle SQL statements stored in a file which you can use to run sequential scripts.

Objects extracted

Metadata Command Center extracts the following metadata from the Oracle Database source system:

- Calculation
- Folder
- Script
- Statements

Data lineage

The following lineage data is available for Calculation assets:

- Data set level

Note: You cannot drill down to view the data element level lineage for the Statement assets. Select a data element and open the **Lineage** tab to view the data element level lineage.

For more information about data lineage, see the *Asset Discovery* help.

Oracle SQL Loader

Oracle SQL Loader is a utility that you can use to bulk load data from external files into Oracle database tables.

Objects extracted

Metadata Command Center extracts the following metadata from an Oracle SQL Loader source system:

- Calculation
- Script
- Statement

Data lineage

The following lineage data is available for Oracle SQL Loader assets:

- Oracle to Oracle
- File to Oracle

For more information about data lineage, see the *Asset Discovery* help.

PostgreSQL

PostgreSQL is an open-source relational database management system (RDBMS) used as a data store or data warehouse for applications.

Objects extracted

Metadata Command Center extracts the following objects from a PostgreSQL source system:

- Database
- Foreign Table

  Note: Objects of the Foreign Table type appear as Table in Data Governance and Catalog.
- Schema
- Table
- View
- Materialized View

  Note: Objects of the Materialized View type appear as View in Data Governance and Catalog.
- Stored Procedure

Data lineage

The following lineage data is available for PostgreSQL assets:

- From table to view

For more information about data lineage, see the *Asset Discovery* help.

Qlik Sense

Qlik Sense is a business intelligence and visual analytics platform that supports a range of use cases including centrally deployed analytic applications, dashboards, and embedded analytics within a scalable framework.

Objects extracted

Metadata Command Center extracts the following objects from a Qlik Sense source system:

- Application
- Application Model
- Column

- Condition
- Connection
- Connection Schema
- Database Schema
- Dimension
- Dimension Attribute
- Document
- Expression
- Field
- File
- File Directory
- Folder
- Hub
- Measure
- Measure Attribute
- QVD
- QVD Model
- Source Column
- Source Table
- Story
- Stream
- System Field
- Table

Relationships

The relationships between the following Qlik Sense assets are derived and displayed:

- Hub to Stream
- Stream to Document

Data lineage

Data lineage is available for the following Qlik Sense assets:

- Document
- Table and its child assets

For more information about data lineage, see the *Asset Discovery* help.

Qlik Sense Cloud

Qlik Sense Cloud is a cloud-based business intelligence tool to connect to data and create reports. It offers modern analytics that enables you to uncover insights and initiate actions.

Objects extracted

Metadata Command Center extracts the following metadata from a Qlik Sense Cloud source system:

- Application
- Application Model
- Column
- Condition
- Dimension
- Dimension Attribute
- Document
- Expression
- Field
- File
- File Directory
- Folder
- Hub
- Measure
- Measure Attribute
- QVD
- QVD Model
- Source Column
- Source Table
- Space
- Story
- System Field
- Table

Data lineage

Data lineage is available for the following Qlik Sense Cloud assets at the data set and data element levels:

- Document
- Table and its child assets

For more information about data lineage, see the *Asset Discovery* help.

QlikView

QlikView is flexible business intelligence tool to connect to data and create reports. Create a QlikView catalog source to extract metadata from the QlikView source system.

Objects extracted

Metadata Command Center extracts the following objects from a QlikView source system:

- Application
- Button Field
- Column
- Condition
- Connection
- Connection Model
- Connection Schema
- Database Schema
- Document
- Expression
- Field
- File
- File Directory
- Folder
- Group
- Measure
- QVD
- QVD Model
- QVW
- QVW Model
- Source Column
- Source File
- Source Table
- System Field
- System Variable
- Table
- Text Field
- Variable

Data lineage

The following lineage data is available for QlikView assets:

- Report level
- Report element level

For more information about data lineage, see the *Asset Discovery* help.

SAP BW/4HANA

SAP BW/4HANA is an enterprise data warehouse platform that captures, stores, and consolidates enterprise data.

The SAP BW/4HANA catalog source extracts metadata from the following assets in SAP BW/4HANA source systems:

- Advanced DSO
- Aggregation Level
- Characteristics
- Composite Provider
- Data Source
- DTP
- Info Area
- Info Objects
- InfoObject as InfoProvider
- Info Source
- Key Figures
- Navigational Attributes
- Open Hub Destination
- Open ODS View
- Source System
- Transformation
- Transformation Rule
- Query
- Query Element or Field

You can extract metadata from SAP BW/4HANA Cloud private edition. This includes an SAP BW/4HANA application hosted in a private cloud environment under a RISE with SAP subscription.

Note: If the virtual machines are provisioned in different subnets, ensure that the necessary ports are open to allow two-way communication between the virtual machines.

Data lineage

The following lineage data is available for SAP BW/4HANA assets:

- Data set level
- Data element level

For more information about data lineage, see the *Asset Discovery* help.

SAP BusinessObjects

SAP BusinessObjects is a business intelligence tool to connect to data for data reporting, visualization, and sharing. Configure an SAP BusinessObjects catalog source to extract metadata from SAP BusinessObjects.

Objects extracted

Metadata Command Center extracts the following objects from an SAP BusinessObjects source system:

- Alias Column
- Alias Table
- Analytic Model
- Analytic Report
- Business Layer View
- Category
- Cell
- Class
- Column
- Connection
- Connection Classifier
- Connection Feature
- Connection Folder
- Connection Model
- Constants
- Context
- Contexts
- Crystal Report
- Crystal Report Model
- Custom Hierarchies
- Dashboard
- Data Provider
- Data Providers
- Database Connection
- Default Hierarchies
- Derived Table
- Desktop Intelligence Document
- Desktop Intelligence Model
- Detail
- Detail Result
- Detail Variable
- Dimension
- Dimension Result
- Dimension Variable
- Discovered Tables
- Enterprise Folder
- Favorite Folder

- Field
- File
- File Connection
- Filter
- Folder
- Formula Fields
- Formulas
- Group Name Fields
- Hierarchy
- Inbox
- Join
- Key
- List of Values
- Logical Column
- Measure
- Measure Result
- Measure Variable
- Merged Dimensions
- Object Package
- Olap Connection
- Olap Dimension
- Parameter
- Parameter Answer
- Parameter Prompt
- Personal Category
- Procedure Query
- Procedure Table
- Query
- Query Filter
- Report Field
- Report Filter
- Report Folder
- Repository
- Running Total Fields
- SQL Expression Fields
- SQL Query
- Schema
- Special Fields
- Summary Fields

- Table
- Universe
- Universe Connection
- Universe Folder
- Universe Model
- Universe Query
- Value
- Variables
- View
- Web Intelligence Document
- Web Intelligence Model
- XML Query

Data lineage

Data lineage is available for the following SAP BusinessObjects assets:

- Analytic Report
- Crystal Reports
- Desktop Intelligence Document
- Web Intelligence Document

For more information about data lineage, see the *Asset Discovery* help.

SAP BusinessObjects Data Services

SAP BusinessObjects Data Services is an ETL tool used for data integration, data quality, data profiling and data processing. It enables you to integrate and transform reliable data into a data warehouse system for analytical reporting.

Objects extracted

Metadata Command Center extracts the following metadata from an SAP BusinessObjects Data Services source system:

- Project
- Job
- Workflow
- Dataflow
- Dataflow Instance
- Script
- Script Instance
- Calculation

Data lineage

Data lineage is available for the following SAP BusinessObjects Data Services assets at the data set and data element levels:

- DataflowInstance
- ScriptInstance
- Calculation

For more information about data lineage, see the *Asset Discovery* help.

SAP Business Warehouse

SAP Business Warehouse (SAP BW) is a data warehouse product of SAP that is based on the SAP NetWeaver ABAP environment. SAP BW can transform and consolidate business information from any source system for reporting, analysis, and interpretation of business data. This data is accessible through built-in reporting, business intelligence and analytics tools as well as third-party software.

Objects extracted

Metadata Command Center extracts the following objects from an SAP BW source system:

- InfoArea
- Data Source
- InfoObject
- InfoObject Catalog
- Source System
- InfoSource
- Characteristic
- Key Figure
- Dimension
- Navigational Attribute
- InfoSet
- DataStore Object (DSO)
- InfoCube
- MultiProvider
- Transformation
- Transformation Rule
- Query
- Query Element or Field
- Workbook
- Advanced DSO
- Composite Provider

- DTP
- InfoObject as InfoProvider
- CDS View
  The catalog source job extracts only CDS views that have queries.
- Open Hub Destination

Note: If an incremental metadata extraction job includes a view, it extracts the corresponding tables even if the tables were extracted in previous runs.

Data profiling for SAP BW objects

Configure data profiling to run profiles on the metadata extracted from an SAP BW source system. You can view the profiling statistics in Data Governance and Catalog.

You can run data profiles on the following SAP BW objects:

- InfoCube
- MultiProvider
- DataStore Object (DSO)

Sampling type

You can run data profiles on all rows in the metadata extracted from SAP BW source systems.

Data lineage

The following lineage data is available for SAP BW assets:

- Data set level
- Data element level

For more information about data lineage, see the *Asset Discovery* help.

SAP Cloud for Customer

SAP Cloud for Customer is an on-demand cloud-based customer relationship management system that helps you manage day-to-day sales contacts.

Objects extracted

Metadata Command Center extracts the following metadata from an SAP Sales & Service Cloud source system:

- Foreign Key
- Primary Key
- SAP Entity
- SAP Entity Field
- SAP Schema
- SAP Service

SAP ERP

SAP Enterprise Resource Planning (ERP) is a platform that incorporates the key business functions of an organization. Use this catalog source to extract metadata from SAP ERP Central Component (ECC) or SAP S/4HANA source systems.

The SAP ERP catalog source extracts metadata from the following assets in SAP S/4HANA and SAP ECC source systems:

- Classes in active state
- Data Elements in active state
- Data Sources built on tables, views and function modules
- Domains in active state
- Executable programs in active state
- Fields
- Packages
  Includes subpackages and assets in subpackages.
- RFC Connections in active state
- RFC function modules and function modules used in a data source
- Tables in active state such as transparent tables, cluster tables, and pool tables
- Transaction codes in active state
- Views such as database views, CDS views, append views, and external views

Note: If an incremental metadata extraction job includes a view, it extracts the corresponding tables even if the tables were extracted in previous runs.

Data classification for SAP ERP objects

Configure data classification for SAP ERP catalog sources to classify and organize data in your organization.

You can choose one of the following options:

- **Data Classification Rules**. Choose from predefined or custom data classifications.
- **Generated Data Classifications**. CLAIRE automatically generates data classifications for the data elements.

  Note: SAP ERP source systems can contain ambiguous data with technical field names. To ensure that CLAIRE generates accurate data classifications, it uses the extracted **Business Name** attribute to generate data classifications.

You can view the data classification results in Data Governance and Catalog.

For more information about data classifications, see *Data classification* in the Administration help.

- Create an SAP Table connection in IDMC Administrator.
- To read data from SAP tables, install the SAP Table connection transport files from the Secure Agent directory to the SAP system.

For information about the configuration of the SAP Table connection in Administrator and the installation of transport files to read from SAP tables, see *SAP Connector* in the Data Integration Connectors help.

Configure data profiling to run profiles on metadata extracted from SAP S/4HANA and SAP ECC source systems.

You can run data profiles on the following SAP ERP objects:

- Table
- View

  Note: You can run data profiles on database views and CDS views.

After you run the catalog source job, you can view the profiling statistics in Data Governance and Catalog.

The data profiling task runs profiles on the following data types for SAP ERP objects:

- ACCP
- CHAR
- CLNT
- CUKY
- CURR
- DATS
- DEC
- D16D
- D16R
- D34D
- D34R
- FLTP
- INT2
- INT4
- LANG
- LCHR
- LRAW
- NUMC
- QUAN
- SSTRING
- STRING
- STRU
- TIMS
- UNIT
- RAW
- RAWSTRING

For information about rules and guidelines for SSTRING, STRING, and RAWSTRING data types, see *SAP Connector* in the Data Integration-Free and PayGo help.

Sampling type

Determine the sample rows on which you want to run a data profiling task.

You can choose one of the following sampling types:

- All Rows

- Limit N Rows

Data lineage

Lineage data is available for SAP ERP assets that connect to the following source systems:

- SAP Business Warehouse (BW)
- SAP BW/4HANA

The lineage for SAP ERP Central Component (ECC) and SAP S/4HANA source systems indicates the tables that relate to each other through primary keys and foreign keys. This is indicated by the labels "PK" and "FK".

For more information about data lineage, see the *Asset Discovery* help.

SAP HANA

SAP HANA Database is a multi-model database system that stores data in its memory and performs high-speed processing and analytics of data.

Objects extracted

Metadata Command Center extracts the following objects from an SAP HANA Database source system:

- Table
- Column
- Plain SQL Database Views (Views created using the CREATE VIEW syntax)
- View Column
- Schema
- Calculation View
  Includes calculation views created in the HANA Deployment Infrastructure (HDI).
- Attribute
- Constraints (Primary Key and Foreign Key)
- Sequence

Data classification for SAP HANA Database objects

Configure data classification for SAP HANA Database catalog sources to classify and organize data in your organization. You can view the data classification results in Data Governance and Catalog.

Data lineage

The following lineage data is available for SAP HANA Database assets:

- From table to view

For more information about data lineage, see the *Asset Discovery* help.

SAP PowerDesigner

SAP PowerDesigner is a collaborative enterprise data modeling tool that enables enterprises to visualize, analyze, and manipulate metadata for effective enterprise information architecture.

Objects extracted

Metadata Command Center extracts the following metadata from an SAP PowerDesigner source system:

- Association
- Business Rule
- Column
- Condition
- Database
- Default
- Dimension
- Dimension Attribute
- Domain
- Extended Attribute
- Extended Attributes
- Fact
- Fact Attribute
- Foreign Key
- Hierarchy
- Index
- Key
- Measure
- Multidimensional Diagram
- Named RelationEnd
- Package
- Physical Data Model
- Physical Diagram
- Primary Key
- Procedure
- Reference
- Schema
- Sequence
- Synonym
- Table
- Trigger
- Type Value
- Unnamed RelationEnd
- User
- View
- View Column
- View Reference

SAP S/4HANA Cloud

SAP S/4HANA Cloud is a software-as-a-service (SaaS) version of the SAP S/4HANA ERP source system. It runs on the SAP S/4HANA in-memory database and provides a single place to access, process, and store enterprise data in real time. The SAP S/4HANA Cloud catalog source connects to SAP S/4HANA Cloud Public Edition sources.

Objects extracted

Metadata Command Center extracts the following metadata from an SAP S/4HANA Cloud Public Edition source system:

- Table
- CDS View
  - View Entity
  - View
  - Custom Entity
- Data Element
- Domain

Data lineage

The following lineage data is available for SAP S/4HANA Cloud assets:

- From table to view
- From view to view
- From unreleased dictionary object to view

For more information about data lineage, see the *Asset Discovery* help.

SAP SuccessFactors

SAP SuccessFactors is a cloud-based software for human capital management (HCM).

Objects extracted

Metadata Command Center extracts the following metadata from an SAP SuccessFactors source system:

- Foreign key
- Primary key
- SAP schema
- SAP table
- SAP table field
- SAP view
- SAP view field

Salesforce

Salesforce is a cloud-based platform that integrates Salesforce CRM, Salesforce Data 360, and Salesforce Flows applications for customer service, analytics, and automation.

Objects extracted

Metadata Command Center extracts metadata of the following objects from a Salesforce CRM application:

- Organization
  - List View
    - List View Column
  - Trigger
  - Salesforce Object
    - Field

Note: If there is a duplicate global list view that is not associated with a specific object in the source system, then only one global list view is extracted.

Metadata Command Center extracts metadata of the following objects from a Salesforce Data 360 application:

- Organization
  - Data Lake Object
    - Data Lake Object Field
  - Data Space
    - Calculated Insight created with Visual Builder or by SQL authoring
      - Dimension
      - Measure
    - Data Mapping linked to Data Model Object
    - Data Model Object mapped and used within a Salesforce Data 360 application
      - Data Model Object Field
    - Identity Resolution
      - Match and Reconciliation
  - Data Stream
    - Data Stream Calculation including Formula Field and Filter
  - Data Transform
    - Batch Data Transform and Streaming Data Transform including Data Transform Definition

Metadata Command Center extracts metadata of the following objects from a Salesforce Flows application:

- Organization
  - Flow
    - Autolaunched Flow (No Trigger)
    - Data 360-Triggered Flow
    - Record-Triggered Flow

- Schedule-Triggered Flow

Note: You can only extract metadata of the unmanaged flows.

Data lineage

The following lineage data is available for Salesforce assets:

- Data set level
- Data element level

You can extract lineage information from the following source systems:

- Salesforce
- Microsoft Azure Blob Storage
- Amazon S3
- Amazon Redshift
- Secure File Transfer
- Snowflake
- Databricks
- Microsoft SQL Server
- Google Big Query
- Workday
- Google Cloud Storage
- Microsoft Azure MySQL
- Microsoft Azure PostgreSQL

Salesforce Marketing Cloud

Salesforce Marketing Cloud is a unified digital marketing automation platform that helps organizations use data and artificial intelligence to reach customers and prospects across channels such as email, web, and social media.

Objects extracted

Metadata Command Center extracts the following metadata from a Salesforce Marketing Cloud source system:

- Attribute Group
- Business Unit
- Campaign
- Data Extension
- Data Extension Attribute
- Folder
- Import Definition
- Import Definition Calculation

- Journey
- Journey Calculation
- Population

Data lineage

The following lineage data is available for Salesforce Marketing Cloud assets:

- Data set level
- Data element level
- Catalog source level

SFTP File System

Description

SFTP is a secure file transfer protocol that enables you to transfer files over SSH, also known as Secure Shell protocol.

Objects extracted

Metadata Command Center extracts metadata from the following objects:

- File Systems
- Flat Fields
- Flat Files
- Folders from an SFTP File System source

Supported file types

You can extract metadata from the following file types:

- Attribute
- AVRO
- CSV
- Element
- JSON
- Markdown
- Microsoft Excel files
  - Excel 97-2003 Workbook with XLS extension
  - Excel Workbook with XLSX extension
  - Excel Macro-Enabled Workbook with XLSM extension
- Microsoft Word files
  - DOC
  - DOCX
- Parquet files based on the partition structure

- PDF
- TSV
- TXT
- XML File
- XSD File

The following table lists the structures associated with the file types that you can extract metadata from:

File Type	Partition structure
AVRO	Single partition, multiple partitions, schema merge
CSV	Single partition, multiple partitions, schema merge
JSON	Single partition, multiple partitions, schema merge
Parquet	Single partition, multiple partitions, schema merge
XML	Single partition, multiple partitions, schema merge

You can extract workbooks, worksheets, and columns from Microsoft Excel files.

You can extract metadata from XML and XSD file formats as XML file objects. Metadata Command Center extracts only elements and attributes from XML and XSD files. If the size of the XML file exceeds 100 KB, Metadata Command Center extracts metadata from the initial 100 KB of the file. However, for XSD file types, Metadata Command Center extracts complete metadata.

Snowflake

Snowflake is an analytic data warehouse provided as Software-as-a-Service (SaaS). The Snowflake data warehouse uses an SQL database engine with a unique architecture designed for cloud services.

Snowflake tags

In Snowflake, you can assign a tag to a database object. When a tag is assigned to an object, the tag is assigned a string value. Snowflake stores the tag and its string value as a key-value pair. For example, `cost_center = 'sales'`. When you run the metadata extraction task for a Snowflake catalog source, the catalog source job extracts Snowflake tags by default. In Data Governance and Catalog, you can view Snowflake tags for data elements and entities in a catalog source and also search assets by Snowflake tags.

For more information about Snowflake tags, see the Snowflake documentation.

Objects extracted

The metadata extraction service extracts the following objects from a Snowflake source system:

- Database
- Schema
- Table
- Tag

- View
- Materialized View

  Note: Objects of the Materialized View type appear as View in Data Governance and Catalog.
- Function

  Note: Objects of the Function type appear as Stored Procedure in Data Governance and Catalog.
- Stored Procedure

  Note: To extract Data Definition Language (DDL) statements from a stored procedure, you need owner privileges on the stored procedure. Alternatively, you can re-create the stored procedure with the `EXECUTE AS CALLER` clause.

  Note: You can extract lineage from procedures and functions in the JavaScript language.
- Pipe
- Stage
- Column

  Note: For certain column types, such as DATE, the column length in bytes is not applicable because the column type has a fixed length.
- External Table

  - Iceberg Table

You can extract metadata from stored procedures that use the following languages:

- JavaScript
- Snowflake SQL scripting
- Snowpark Python

Note: Effective in the 2024.11.S release, extracting metadata from stored procedures that use Snowpark Python is available for preview.

Preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.

Secure Data Sharing in Snowflake allows you to share selected database assets with other Snowflake accounts without copying or transferring any actual data. To share data, providers create a share of their database, select specific objects to include, and then add consumer accounts to this share.

You can extract metadata from the following database assets shared through Snowflake Secure Data Sharing:

- Database
- Table
- External table
- Secure view
- Secure materialized view

To view the complete data lineage and all the metadata extracted for the consumer account, including shared assets, perform a connection assignment between catalog sources created for the provider and consumer Snowflake accounts.

Data profiling for Snowflake objects

Configure data profiling to run profiles on the metadata extracted from a Snowflake source system. You can run data profiles on the following Snowflake objects:

- Table
- View

You can view the profiling statistics in Data Governance and Catalog. The data profiling task runs profiles on the following data types for Snowflake objects:

- NUMBER
- FLOAT
- DOUBLE
- VARCHAR
- BOOLEAN
- DATE
- TIME
- TIMESTAMP_LTZ
- TIMESTAMP_NTZ
- TIMESTAMP_TZ
- OBJECT
- ARRAY
- VARIANT

Sampling type

Determine the sample rows on which you want to run the data profiling task. You can choose one of the following sampling types for a Snowflake catalog source:

- All Rows
- Limit N Rows
- Custom Query

Data classification for Snowflake objects

You can run the data classification capability on Snowflake catalog sources to identify and organize data into relevant categories. For more information about data classification, see the *Administration* help.

Data lineage

The following lineage data is available for Snowflake assets:

- From table to view
- Stored Procedures linked to tables or views
- Functions linked to tables or views
- Procedure calling a procedure
- Procedure calling a function
- Pipe linked to stages
- Pipe linked to table or views

For more information about data lineage, see the *Asset Discovery* help.

Snowflake SQL Script

{Varref: Snowflake SQL Script}Snowflake SQL Script
is a set of Snowflake SQL statements stored in files that you can use for sequential execution.

Objects extracted

Metadata Command Center extracts the following metadata from the Snowflake SQL Script source system:

- Calculation
- Folder
- Script
- Statements

Data lineage

The following lineage data is available for Statements and Calculation assets:

- Data set level
- Data element level

Note: You cannot drill down to view the data element level lineage for the Statement assets. Select a data element and open the **Lineage** tab to view the data element level lineage.

For more information about data lineage, see the *Asset Discovery* help.

Swagger API

Swagger API is a set of open-source tools built around the OpenAPI specification that help you design, build, document, and consume REST APIs.

Objects extracted

Metadata Command Center extracts the following metadata from a Swagger API source system:

- Operation
- Parameter
- Path
- Property
- RequestBody
- Response
- Schema
- Security
- SwaggerAPI
- Tag

The Swagger API catalog source works with version 3.x.x of the OpenAPI specification.

Strategy Cloud

Strategy Cloud is an AI-powered, cloud-native business intelligence platform designed to help organizations build, explore, and act on data.

Objects extracted

Metadata Command Center extracts the following metadata from a Strategy Cloud source system:

- Attribute
- Attribute Form
- Column
- Dashboard
- Datamart Report
- Document
- Fact
- Fact Measure
- Folder
- Hidden Folder
- Logical Table
- Logical Table Attribute Form
- Logical Table Fact Measure
- Metric
- Metric Measure
- Project
- Report
- Report Attribute
- Report Attribute Form
- Report Metric
- Report Metric Measure
- Warehouse Table

Data lineage

The following lineage data is available for Strategy Cloud assets:

- Data element level
- Data set level

Note: Strategy Cloud assets show the same lineage information at data element and data set level.

For more information about data lineage, see the *Asset Discovery* help.

Tableau

Tableau is a business intelligence tool that connects to data and create dashboards that can be shared.

Objects extracted

Metadata Command Center extracts the following objects from a Tableau source system:

- Calculation
- Data Source
- Dashboard
- Workbook
- Worksheet
  - Measure Names

    Note: Objects of the measure name type appear as calculations nested under worksheets in Data Governance and Catalog.
- Project
- Server
- Site
- Stored Procedure
- Tags

Data lineage

Lineage data is available for Tableau assets that connect to the following source systems:

- Amazon S3
- Microsoft Azure Data Lake Storage Gen2
- Microsoft Azure SQL Server
- Oracle
- Snowflake
- IBM Netezza using JDBC
- Microsoft SQL Server
- IBM DB2
- Teradata Database
- Amazon Redshift
- Snowflake
- Apache Hive
- Google BigQuery

The following lineage data is available for Data Source and Worksheet assets:

- Data set level
- Data element level

The data set level lineage is available for Dashboard asset and data element level lineage is available for data element level.

For more information about data lineage, see the *Asset Discovery* help.

TIBCO Spotfire

TIBCO Spotfire is a business intelligence tool that you can use to explore data through dashboards and advanced analytics.

Objects extracted

Metadata Command Center extracts the following metadata from the TIBCO Spotfire source system:

- AnalysisDocument
- Annotation
- Column
- ColumnMatch
- ConnectionClassifier
- ConnectionDataSource
- ConnectionFeature
- CustomQuery
- DataColumn
- DataFunction
- DataProducer
- DataTable
- DataTransformation
- Database
- Document
- DocumentProperty
- ESRIShapeFile
- ExcelWorkbook
- ExcelWorksheet
- Expression
- Field
- File
- Filter
- Folder
- InformationLink
- LocalConnection
- Parameter
- Property
- Relation
- Report
- ResultAttribute
- SBDFFile
- STDFFile

- Schema
- Server
- StoredProcedure
- Table
- TextFile
- View

Data lineage

Data lineage is available for the following TIBCO Spotfire assets:

- Column
- Data Column
- Data Producer
- Data Table
- Field
- Report
- Schema
- Table
- View

For more information about data lineage, see the *Asset Discovery* help.

Talend Data Integration

Talend Data Integration is an ETL data integration platform that you can use to gain business insights from data.

Objects extracted

Metadata Command Center extracts the following metadata from the source system:

- Project
- Folder
- Job
- Subjob
- Calculation

Metadata Command Center extracts the following metadata if the source system uses stored procedures for connection-aware scans:

- Project
- Folder
- Job
- Subjob
- Calculation

- Database
- Procedure Definition
- Procedure Instance
- Parameter
- Parameter Container
- Schema
- Statement

Data lineage

The following lineage data is available for Calculation and Subjob assets:

- Data set level
- Data element level

For more information about data lineage, see the *Asset Discovery* help.

Teradata BTEQ Script

Teradata BTEQ Script is a set of commands and Teradata SQL statements stored in files that you can use to run sequential scripts.

Objects extracted

Metadata Command Center extracts the following metadata from the Teradata database source system:

- Calculation
- Folder
- Script
- Statements

Data lineage

The following lineage data is available for Statements and Calculation assets:

- Data set level
- Data element level

Note: You cannot drill down to view the data element level lineage for the Statement assets. Select a data element and open the **Lineage** tab to view the data element level lineage.

For more information about data lineage, see the *Asset Discovery* help.

Teradata FastExport Script

Teradata FastExport Script is a set of Teradata FastExport statements stored in a file that you can use to transfer large amounts of data from tables and views of the Teradata Database to a client-based application.

Objects extracted

Metadata Command Center extracts the following metadata from the Teradata database source system:

- Calculation
- Folder
- Script
- Statements

Data lineage

The following lineage data is available for Teradata FastExport Script assets:

- Data set level
- Data element level

For more information about data lineage, see the *Asset Discovery* help.

Teradata FastLoad Script

Teradata FastLoad Script is a utility that you can use to quickly load large amounts of data in an empty table on a Teradata database.

Objects extracted

Metadata Command Center extracts the following metadata from a Teradata database source system:

- Calculation
- Folder
- Script
- Statement

Data lineage

The following lineage data is available for Teradata FastLoad Script assets:

- Data set level
- Data element level

For more information about data lineage, see the *Asset Discovery* help.

Teradata MultiLoad Script

Teradata MultiLoad Script is a set of Teradata MultiLoad statements stored in a file that you can use to import data from input files, applications or other correctly formatted sources to a Teradata Database.

Objects extracted

Metadata Command Center extracts the following metadata from the Teradata database source system:

- Calculation

- Folder
- Script
- Statements

Data lineage

The following lineage data is available for Teradata MultiLoad Script assets:

- Data set level
- Data element level

For more information about data lineage, see the *Asset Discovery* help.

Teradata Database

Teradata Database is a Relational Database Management System (RDBMS) solution for data warehousing applications.

Objects extracted

Metadata Command Center extracts the following objects from a Teradata Database source system:

- Database
- Schema. The service extracts schemas along with descriptions.
- Table
- View
- Column
- Stored Procedure

Workday

Workday is a cloud-based financial management, enterprise resource planning, and human capital management platform.

Extracted objects

Metadata Command Center extracts the following metadata from a Workday source system:

- Business Object
- Data Source
- Operation
- Operation Entity
- Operation Entity Property
- Report
- Report Entity

- Report Entity Property
- Report Property
- Service

Data lineage

The following lineage data is available for Workday assets at the data set level:

- From Business Object to Data Source
- From Data Source to Report

For more information about data lineage, see the *Asset Discovery* help.

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